Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends*

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Abstract

Cross-sectional firm-size dispersion reflects how average firm size evolves with age, the firm-age distribution, and size heterogeneity conditional on age. Using high-quality Swedish administrative data, I document a steepening of the average firm-size trajectory. I develop a creative-destruction model in which entrants—differing in productivity—compete with incumbents by expanding into new product markets. While rising entry costs shift the firm-age distribution and greater productivity differences raise within-age size heterogeneity, declining expansion costs increase cross-sectional dispersion by steepening the average firm-size trajectory. A structural estimation identifies declining expansion costs as the primary driver of the rise in firm-size dispersion.

Keywords: Firm dynamics, Long-run macroeconomic trends, Administrative data

JEL codes: D22, E24, O31, O47, O50

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1 Introduction

Over recent decades, advanced economies have exhibited a common macroeconomic trend: large incumbent firms have grown disproportionately relative to smaller firms, leading to a pronounced rise in firm-size dispersion.[1] The underlying causes of this trend remain widely debated, with explanations ranging from growing productivity advantages of 'superstar' firms (Autor et al., 2020; Aghion et al., 2023; De Ridder, 2024) to rising entry barriers (Davis, 2017; Gutiérrez and Philippon, 2018). This paper contributes to this debate in three ways. First, I document systematic changes in firm-size dynamics and show, using a simple decomposition, how these changes map into rising cross-sectional size dispersion. Second, I develop a model of creative destruction that extends existing frameworks to allow a unified analysis of how the forces emphasized in the debate shape firm-size dispersion. Third, leveraging the empirical patterns documented in the first part, I use the model to conduct a quantitative horse race that identifies the contribution of each force to the observed rise in firm-size dispersion.

The paper begins with a simple decomposition that relates cross-sectional firm-size dispersion to firm-size dynamics. Dispersion increases when the average size of surviving firms rises more steeply with age, when the firm-age distribution shifts toward older (larger) firms, or when residual size heterogeneity conditional on age expands. Using high-quality administrative tax records covering the universe of Swedish firms from 1997 to 2017, I show that the average size–age trajectory has steepened—corresponding to the first component in the decomposition. For example, among firms founded in the 1990s, real value added at age eight was approximately 49 percent higher than at entry; for those founded in the 2010s, the increase was about 67 percent. A similar pattern emerges for the wage bill: among firms founded in the 1990s, the wage bill at age eight was about 47 percent higher than at entry, compared to 63 percent for firms founded in the 2010s. Alongside the change in the average size trajectory, I document a marked increase in cross-sectional firm-size dispersion using the same data, as well as other macroeconomic trends—particularly declining firm entry and slowing productivity growth.

Firm-size trajectories are endogenous equilibrium outcomes shaped by firms' dynamic decisions. I develop a structural model that endogenizes these trajectories to identify the forces driving the documented trends. At its core, the model is an endogenous-growth framework of creative destruction in the spirit of the quality-ladder models of Aghion and Howitt (1992) and Klette and Kortum (2004). Incumbent firms and potential entrants expand into new markets by improving the quality of competitors' products through innovation (expansion R&D). Such models have been shown to capture firm dynamics well (Lentz and Mortensen, 2008; Akcigit and Kerr, 2018). I extend the Klette-Kortum model by introducing ex-ante heterogeneity in firm productivity: entrants differ exogenously in their production efficiency, with some systematically more efficient than others. I use this framework to analyze how different forces, particularly increasing productivity dispersion, rising entry barriers and de-

[1]See, among others, Akcigit and Ates (2021); Autor et al. (2020); Grullon et al. (2019); Decker et al. (2016, 2020); Gourio et al. (2014); Chen et al. (2023).

clining expansion costs alter the average firm-size trajectory, size heterogeneity conditional on age, and the firm-age distribution, and thereby shape cross-sectional size dispersion.

I estimate the model first along a balanced growth path, targeting the average firm-size trajectory for cohorts founded in the 1990s, as well as cross-sectional size dispersion, firm entry, and aggregate productivity growth at that time. I then use simple comparative statics to trace how each mechanism affects size dispersion. Rising entry costs reduce the entry rate, shifting the firm-age distribution toward older firms, which are larger on average and exhibit greater size heterogeneity conditional on age. This shift dominates the contributions from changes in the average size trajectory or dispersion conditional on age, indicating that higher entry barriers primarily increase total size dispersion by altering the age distribution. In contrast, rising productivity dispersion generates greater heterogeneity in firms' expansion rates, thereby increasing size differences conditional on age while the average size trajectory is less affected. Finally, declining expansion costs raise expansion rates across firms regardless of their productivity, increasing size dispersion mainly by steepening the average firm-size trajectory.

To quantify which of the mechanisms drive the observed rise in size dispersion, I re-estimate the model along a new balanced growth path. The estimation jointly targets the steepening of the average firm-size trajectory, the rise in cross-sectional firm-size dispersion, the decline in firm entry, and the slowdown in productivity growth. Alongside other forces, it identifies a decline in firm expansion costs. Implementing the identified parameter changes one at a time highlights that this decline alone accounts for both the observed steepening of the average size trajectory and the rise in overall dispersion—consistent with the earlier comparative-statics result that lower expansion costs raise dispersion primarily through steepening the average size trajectory. The other identified parameter changes, which raise dispersion mainly by shifting the firm age distribution or through greater size heterogeneity conditional on age, contribute to a lesser extent to the total increase. A corresponding decomposition of the total change in dispersion across balanced growth paths into its three endogenous components indicates that the largest share (34.4%) is attributable to changes in the average firm-size trajectory, followed by shifts in the age distribution (28.5%) and by movements in dispersion conditional on age (26.2%).[2] Beside its effect on total size dispersion, the decline in expansion costs contributes to the fall in the entry rate, as higher expansion rates require a lower equilibrium entry rate to clear the labor market. However, the increase in expansion rates raise long-run productivity growth, which the estimation counters by lowering the step size of quality improvements.

Lastly, I solve for the transition path toward the new balanced growth path to quantify the implications for welfare. Following the introduction of the estimated parameter changes as shocks starting from the initial balanced growth path, aggregate productivity soars for about six years, boosted by the increase in firms' expansion rates, after which it falls below the initial balanced growth path trend. On net, the fall in long-run growth dominates the short-run boost, resulting in a welfare loss of 4.95 percent in permanent consumption units.

[2] These shares do not sum to 100% because the decomposition is not additive.

I provide further extensions to both the empirical and model results. Although prior work using U.S. Census data finds that firm size conditional on age has remained stable (Hopenhayn et al., 2022; Karahan et al., 2024), I show that the average firm-size trajectory has steepened in several U.S. service sectors. One reason these patterns appear at the aggregate level in Sweden but not in the United States is that a larger share of firms operate in these sectors in Sweden. On the model side, I further decompose the steepening of the average firm-size trajectory into changes in size growth conditional on survival and changes in the composition of surviving firms. The model shows that more productive firms now not only grow faster conditional on survival, but also account for a larger share among survivors, both of which contribute to the steeper average size trajectory.

Further Literature. This paper relates to the literature that studies causes behind recent macroeconomic trends. I contribute to this literature using a novel observation, namely the steepening of the average firm-size trajectory, to test different theories for rising firm-size dispersion. Two further explanations in the literature that are not directly modeled in the framework in this paper highlight the increasing importance of intangible capital in firms' production (De Ridder, 2024; Chiavari and Goraya, 2020; Weiss, 2019) and changes in population growth (Hopenhayn et al., 2022; Karahan et al., 2024; Peters and Walsh, 2021). First, I find that, at least in the Swedish data, the steepening of the firm-size trajectory is not associated with faster capital accumulation, even when including intangible assets in the capital measure. Arguably, intangible assets in balance sheet data are an imperfect measure of intangible capital, but I take this evidence as motivation not to include intangible capital in the model. Although the model abstracts from intangible capital, the structural estimation assigns only a limited role to increasing productivity (or marginal cost) dispersion—one of the primary channels through which intangible capital would be expected to operate. Second, in models of firm dynamics with population growth, the entry margin generally absorbs changes in population growth, shielding firms' size growth from changes in labor supply growth.[3] Engbom (2023) further shows that the growth rate of the workforce remained fairly stable in Sweden during the 1990s and 2000s.

Other studies have embedded ex-ante firm heterogeneity into models of creative destruction. Aghion et al. (2023) develop a model with ex-ante productivity differences, but abstract from firm entry and exit, which are crucial for analyzing the firm size–age relationship. Ex-ante heterogeneity has also been introduced along other dimensions—for example, in innovation step sizes (Lentz and Mortensen, 2008). Heterogeneity in productivity is conceptually distinct: firms must track the entire distribution of productivity types across product lines to form markup expectations, which determine their expansion incentives. This introduces an additional state variable—the productivity distribution—into the firm's optimization problem. By contrast, with heterogeneity in step sizes or R&D costs, firms only need to track their own type.

[3] In Hopenhayn-models, the free entry condition holds the wage constant, rendering firms' labor demand, and hence size trajectories, invariant to population changes (Hopenhayn et al., 2022; Karahan et al., 2024). Also in models of creative destruction, the free entry condition pins down the value of a product line, isolating incumbents' expansion R&D incentives from changes in population growth (Peters and Walsh, 2021).

The paper proceeds as follows. Section 2 introduces a simple decomposition that links firm-size dynamics to cross-sectional size dispersion. Section 3 documents empirical trends in firm-size dynamics and size dispersion. Section 4 presents the model, which I estimate on separate balanced growth paths in Section 5 to evaluate competing explanations for the observed trends. Section 6 analyzes the transitional dynamics. Section 7 provides extensions, and Section 8 concludes.

2 A model-free decomposition of firm-size dispersion

To guide the analysis of firm-size dynamics and firm-size dispersion in the next sections, I begin with a model-free decomposition that links both. Consider the law of total variance, $\text{Var}(Y) = \text{Var}(E[Y|X]) + E[\text{Var}(Y|X)]$, which decomposes the total dispersion in Y into (i) dispersion between groups defined by X and (ii) residual dispersion within those groups. Setting $Y \equiv \ln s_f$ and $X \equiv a_f$, where s_f and a_f denote firm size and age, respectively, this decomposition partitions the cross-sectional dispersion in log firm size into variation in average size across firm ages and residual heterogeneity in size conditional on age.[4] Specifically,

$$\begin{aligned}\text{Var}\left(\ln s_f\right) = &\sum_{a_f} p(a_f)\left[E[\ln s_f|a_f] - E[\ln s_f|0] - \sum_{a_f} p(a_f)\left(E[\ln s_f|a_f] - E[\ln s_f|0]\right)\right]^2 \\ &+ \sum_{a_f} p(a_f)\text{Var}(\ln s_f|a_f), \end{aligned} \tag{1}$$

where $p(a_f)$ is the share of firms of age a_f. In the first term, I add and subtract the average log size of entrants, $E[\ln s_f|0]$, to express between-age variation in terms of deviations from entry size.

Although the decomposition is specific to the variance, it more generally highlights that cross-sectional firm-size dispersion is shaped by three components: (i) the trajectory of average firm size conditional on survival, $E[\ln s_f|a_f] - E[\ln s_f|0]$ for $a_f \in \{0, 1, 2 \dots\}$; (ii) the age distribution of firms, $p(a_f)$; and (iii) residual size heterogeneity within age groups, $\text{Var}(\ln s_f|a_f)$. Total dispersion can increase for three reasons: the average size trajectory may steepen, firm sizes within age brackets may become more dispersed, or the age distribution may shift toward older firms, which are typically larger and exhibit greater within-age dispersion. Eq. (1) provides a natural way to decompose changes in cross-sectional size dispersion into these three components. Empirically, however, implementing this decomposition is challenging because firm-age distributions in available data are typically truncated. I therefore begin by documenting trends in total size dispersion and in firm-size trajectories (for the age groups that can be observed). In later sections, I employ a structural model that permits a full decomposition of overall size dispersion into its three constituent parts. The model allows me to examine how different economic forces affect each component separately

[4] While I use the variance to derive a tractable analytical expression, the underlying intuition extends to alternative dispersion measures.

and, ultimately, to identify which component accounts for the observed rise in cross-sectional size dispersion.

3 Trends in the Swedish economy

Before documenting the empirical trends, I describe the data first.

3.1 Data

Data is provided by Statistics Sweden, the official statistical agency in Sweden. The main data set is *Företagens Ekonomi*, which covers information from balance sheets and profit and loss statements for the universe of Swedish firms at an annual frequency covering the period 1997-2017. If not mentioned otherwise, I focus on the unbalanced panel of firms. The data contains two measures of firm size that I use throughout the paper: value added and the wage bill. For a detailed description of the variables, see Section A in the Appendix. The data further contains information on the firm's industry and legal type. I restrict the analysis to firms operating in the private economy. The firm's birth year is defined as the year the first worker is associated with a firm (employed or self-employed). I obtain this information from the auxiliary data set *Registerbaserad Arbetsmarknadsstatistik*, containing the universe of employer-employee matches. Firms entering late during the year compare small, on average, to firms entering early. To prevent the timing of entry from affecting the average size of entrants within a year, I define the latter as firms aged zero or one. I further restrict the analysis to firms that eventually hire at least one employee, which filters out never-employing firms.

Starting in 2004, the data includes unincorporated self-employed with negative profits. These firms were previously not recorded, leading to an inflow of new firms in that year. The firm entry rate in 2004 is hardly distinguishable from its trend in the cleaned data, suggesting that the stock of these firms is very small. Nevertheless, since I cannot differentiate between firms established in 2004 and firms entering the data in 2004 with an earlier birth year, I drop the cohort of firms entering the data in 2004. For the results of the paper, the change in the data is inconsequential, as the trends I document mostly occur before 2004. Further, the characteristics of entrants have been very stable over time, as shown below, indicating no systematic changes in the type of firms entering the data.

Table 1 reports distributional statistics of firm sales, value added, and production inputs in the data (1997 to 2017). Throughout the paper, nominal variables are deflated to 2017 Swedish Krona (SEK) using the GDP deflator. The median firm lists sales of roughly 1.4 million SEK (approx. 0.14 million US dollars) and employs one full-time worker. The distribution of sales, value added, and all production inputs is highly right-skewed. Mean sales (18.7 million SEK) and full-time employment (6.6) far exceed their respective 75th percentiles. In total, the data includes about 7.4 million firm-year observations. For the firm age- and size-specific empirical analysis, I filter value added at its 0.5% tails and focus

Table 1: Summary statistics (1997-2017)

	25th Pct.	50th Pct.	75th Pct.	Mean	SD	Obs.
*Sales**	0.5	1.4	4.4	18.7	464.8	7,374,865
*Value added**	0.2	0.6	1.7	5.1	116.4	7,374,865
Employment (full-time units)	0	1	3	6.6	107.3	7,374,865
*Wage bill**	0.002	0.24	0.87	2.45	43.4	7,374,865
*Capital stock**	0.02	0.16	0.8	6.9	228.5	7,374,865
*Intermediate Inputs**	0.2	0.5	1.6	7.3	220.8	7,374,865

Note: variables marked with * are in units of million 2017-SEK (1 SEK ≈ 0.1 US dollars). The capital stock is defined as fixed assets minus depreciation.

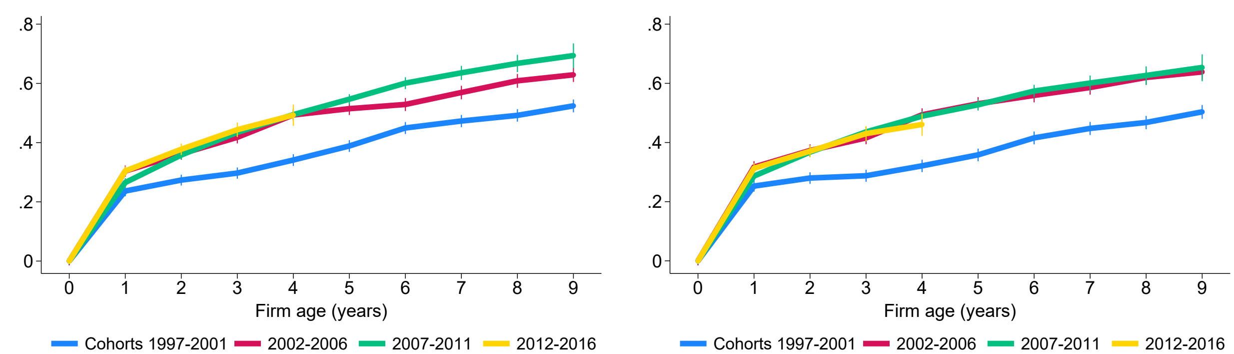
on firms established in 1997 or later, which reduces the sample size to 3.5 million firm-year observations.[5] For firms established in 1997 and later, age is not truncated.

3.2 Trends in the dynamics of firm size

This section documents changes in the average size trajectory of firms. As a starting point, I nonparametrically characterize the relationship between firm size and firm age. Figure 1 plots average firm size as a function of firm age, conditional on survival. Firm size at each age is normalized by the size of entrants, and the relative size is presented in logs. I refer to this measure as firm size relative to entry throughout the remainder of the paper. Panel (a) uses value added to measure firm size, while Panel (b) uses the wage bill. Both panels display 95% confidence intervals around the estimated means.

Figure 1: Firm size relative to entry (by cohort)

(a) Value added (b) Wage bill



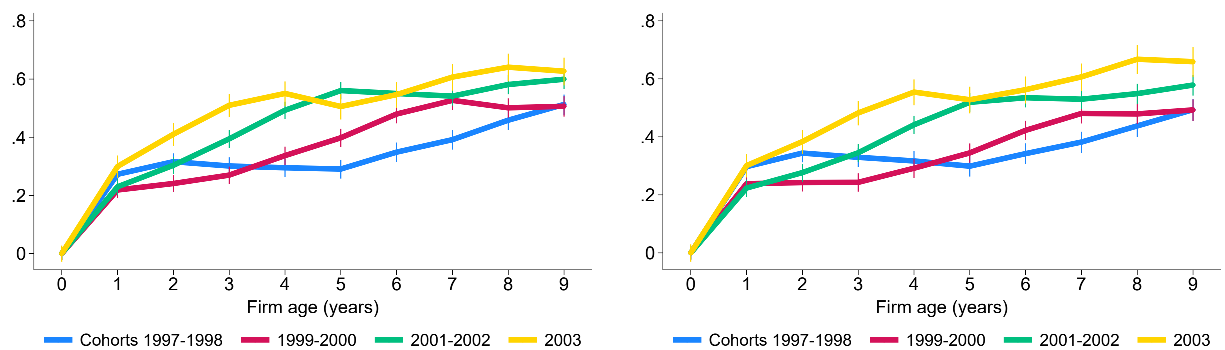
Notes: the figure shows the average value added (left) and wage bill (right) as a function of firm age (conditional on survival) relative to that of entrants in logs for different cohorts as indicated in the legend. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are shown.

[5] I apply the 0.5% filter also to the wage bill and the capital stock, but only at the right tail. This ensures that self-employed without labor costs or firms without any physical capital are included in the analysis.

As expected, firm size increases with firm age; that is, older firms are, on average, larger than younger ones. More notably, however, the relationship between age and size has shifted over time. Figure 1 presents the firm size trajectories separately for the cohorts 1997–2001, 2002–2006, 2007–2011, and 2012–2016. Compared to the initial cohorts (1997–2001), firm size relative to entry has increased for any firm age across all subsequent cohorts. For example, by age eight, value added is 0.492 log points higher than at startup for the 1997–2001 cohorts, compared with 0.668 log points for the 2007–2011 cohorts. Similarly, the wage bill at age eight is 0.468 log points above startup levels for the 1997–2001 cohorts, compared with 0.627 log points for the 2007–2011 cohorts.

Figure 2: Firm size relative to entry (cohorts 1997–2003)

(a) Value added (b) Wage bill

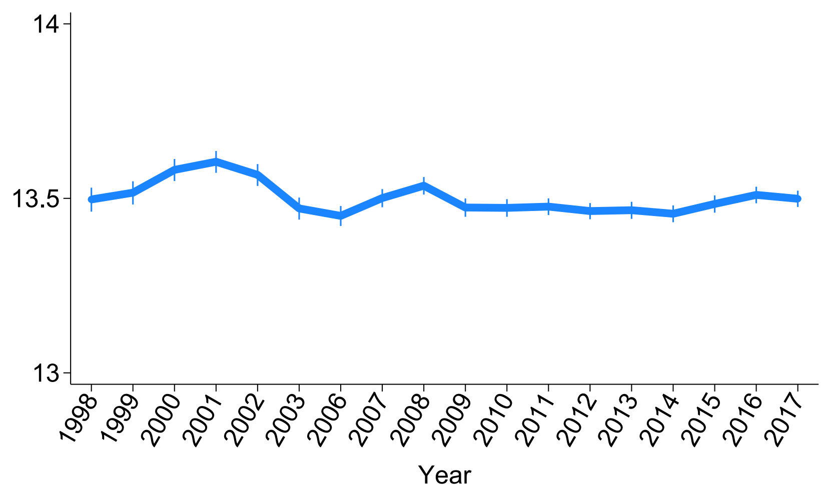


Notes: the figure shows the average value added (left) and wage bill (right) as a function of firm age (conditional on survival) relative to that of entrants in logs for different cohorts as indicated in the legend. 95% confidence bands are included.

Figure 1 indicates that the increase in firm size relative to entry was concentrated primarily among cohorts from the late 1990s and early 2000s. To examine this pattern in greater detail, I focus specifically on these cohorts. Figure 2 presents firm size trajectories for the 1997–1998, 1999–2000, 2001–2002, and 2003 cohorts individually. The figure shows that the increase in size trajectories occurred gradually over this time period. Compared to Figure 1, the trajectories appear less smooth because the averages are calculated over two cohorts rather than five. Nonetheless, a gradual upward shift is evident for both value added and the wage bill. In fact, the size trajectory of the 2003 cohort closely resembles that of the later cohorts shown in Figure 1. Note that the size trajectories in Figures 1 and 2 are conditional on survival and therefore do not necessarily reflect faster firm growth but also potential changes in the selection of surviving firms, a point to which I return later.

Firm size relative to entry has systematically increased over time, while the size of entrants has remained remarkably stable. Figure 3 plots the log of the average wage bill for entrant firms. Firm size at entry is highly stable, particularly over the long run. This stability suggests that there have been no systematic changes in the nature of firms entering the data over time. The years 2004 and 2005 are omitted from the graph because, as noted in the previous section, the 2004 entrant cohort is excluded, and entrants are defined as firms that are zero or one year old. For clarity, all year labels are shown.

Figure 3: Firm size at entry



Notes: the figure shows the average firm size (wage bill) at entry in logs. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one. 95% confidence bands are included.

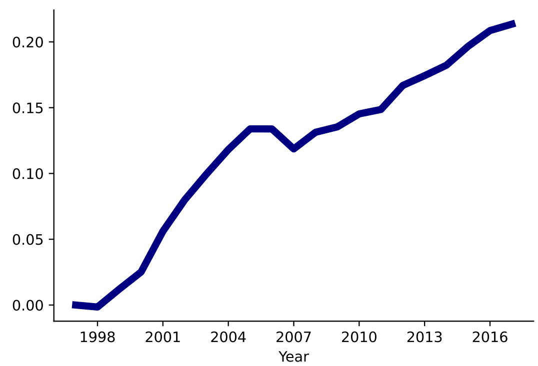
I provide additional robustness checks in Appendix B.1. For instance, Figure A-1a displays firms' physical capital stock relative to entry across all cohorts. The trajectories are nearly identical across cohorts, indicating no systematic differences in capital accumulation over time. This suggests that the observed increase in firm size relative to entry is not driven by capital deepening. This conclusion holds even when adding intangible assets to the capital stock measure, as shown in Figure A-1b. I also display the firm-size trajectories by sector in Figure A-2. The increase in firm size relative to entry is evident in both goods production and services, though it is particularly pronounced among services firms. This pattern implies that the results are unlikely to be driven by external factors, such as foreign demand for Swedish goods. Lastly, I confirm the steepening of the size-age relationship using a regression framework, controlling for cohort and narrowly defined industry fixed effects.

3.3 Trends in the macroeconomy

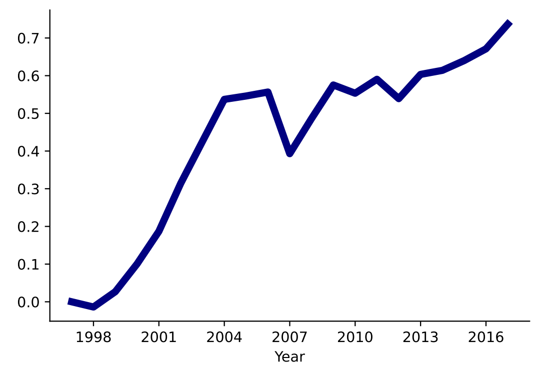
Using the same data, this section documents systematic macroeconomic trends, beginning with rising firm-size dispersion. Figure 4 shows the evolution of two cross-sectional dispersion measures for log value added: the standard deviation and the interquartile range (defined as the difference between the 75th and 25th percentiles). Both measures are calculated within two-digit industries, aggregated using industry labor costs, and normalized to zero in 1997. For the standard deviation (Figure 4a), the sample is restricted to firms with positive value added, as the logarithm is otherwise undefined. For the preferred measure—the interquartile range in Figure 4b—this restriction is not imposed, since the 25th percentile within two-digit industries is typically positive.[6]

Figure 4: Cross-sectional firm size dispersion

(a) Standard deviation



(b) Interquartile range



Notes: the left panel shows the standard deviation and the right panel the interquartile range of log value added. Both measures are computed within two-digit industries and aggregated using industry labor costs. The value in 1997 is normalized to zero.

The interquartile range reveals a pronounced increase in size dispersion over the period 1997–2017: the 75th percentile rose by 0.737 log points relative to the 25th percentile. This implies that, had the two percentiles been equal in 1997, firms at the 75th percentile would be approximately $e^{0.737} \approx 2.09$ times larger than those at the 25th percentile by 2017. Notably, a significant portion of this increase—over 0.5 log points—occurred during the early 2000s. The rise in the interquartile range is more pronounced than the increase in the standard deviation—suggesting that dispersion grew more in the center of the distribution than in the tails—though this comparison should be interpreted with caution given the restriction of the standard deviation to firms with positive value added.

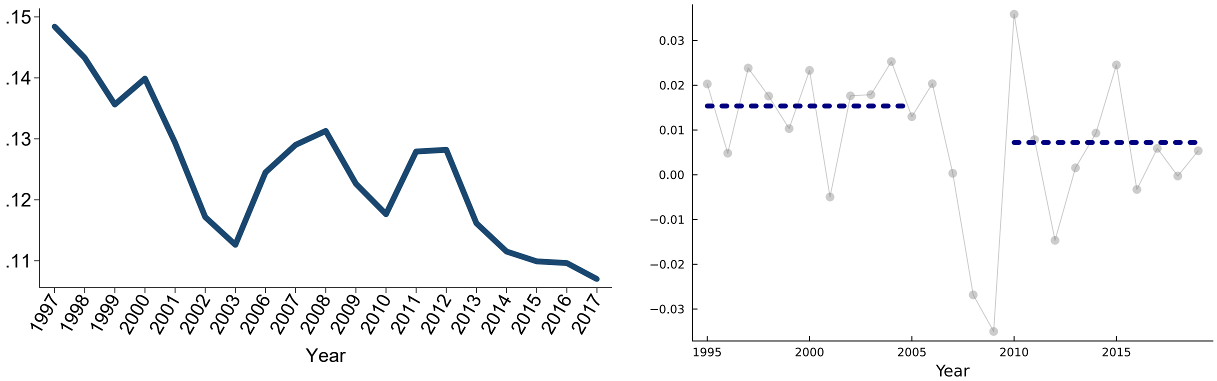
The rise in firm-size dispersion occurred alongside other macroeconomic trends that are common across developed economies. Figure 5a shows the evolution of the firm entry rate between 1997 and 2017, measured as the number of entrants relative to the total number of firms. Over this period, the entry rate declined from about 15% in 1997 to 11% in 2017. Notably, most of this four-percentage-point drop took place around the turn of the millennium. Although the entry rate experienced a modest rebound following the Great Recession, it soon returned to—and even fell below—its earlier levels.[7] As before, the years 2003 and 2004 are excluded, and all year labels are shown in the figure for clarity.

[6]The results are very similar when using finer, five-digit industry classifications, as shown in Appendix B.2. However, this greater granularity increases the frequency of negative values at the 25th percentile of value added, which makes the interquartile range undefined for those product groups.

[7]Engbom (2023) reports a similar decline in the firm entry rate for Sweden.

Figure 5: Other macroeconomic trends

(a) Entry rate (b) Total Factor Productivity growth



Notes: the left panel shows the entry rate defined as the total number of entrants divided by the total number of firms. The years 2004 and 2005 are not shown as I exclude the cohort of firms entering in 2004 and entrants are defined as firms aged zero or one. Right panel: the scatter points in gray show annual Total Factor Productivity (TFP) growth from Federal Reserve Economic Data (FRED). The dashed blue line indicates the respective averages over 1995–2005 and 2010–2019.

Figure 5b presents annual Total Factor Productivity (TFP) growth obtained from Federal Reserve Economic Data (FRED). The series is plotted as scatter points, with the dashed blue line indicating time-averaged values. Sweden experienced strong TFP growth from the mid-1990s through the early 2000s, averaging 1.5% annually between 1995 and 2005. Following the early 2000s, growth slowed, with TFP increasing at an average annual rate of around 1.1% between 2010 and 2015. This pattern—a period of rapid growth around the turn of the millennium followed by a slowdown during the 2010s—mirrors trends observed in many advanced economies, including the U.S. (Aghion et al., 2023).

4 Model

The decomposition in Section 2 links firm-size trajectories to cross-sectional size dispersion, but takes the former as given. This section outlines a rich model of creative destruction that endogenizes firm growth and survival, allowing one to identify the forces that jointly drive the changes in firm-size trajectories and cross-sectional size dispersion.

4.1 Preferences and aggregate economy

Time is continuous and indexed by t. The economy consists of a representative household that chooses the path of consumption C_t and wealth A_t to maximize lifetime utility

$$U = \int_0^\infty \exp(-\rho t) \ln C_t dt,$$

subject to the budget constraint $\dot{A}_t = r_t A_t + w_t L_t - C_t$. ρ denotes the discount rate, r_t the interest rate and w_t the real wage. The household supplies one unit of labor inelastically, i.e., $L_t = 1$. The optimality condition (Euler equation) for the household problem reads

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

Aggregate output is produced competitively using a Cobb-Douglas technology over a continuum of different products indexed by i (t subscripts are suppressed when convenient)

$$Y = \exp\left(\int_0^1 \ln\left[q_i y_i\right] di\right),$$

where y_i and q_i denote the quantity and quality of product i. Output is consumed entirely such that $Y = C$. Expenditure minimization leads to the standard demand function

$$y_i = \frac{YP}{p_i}. \tag{2}$$

P is defined as the aggregate price index, which I normalize to 1.

4.2 Production

Firms, indexed by f, produce in product market i with a linear technology

$$y_{ift} = \varphi_f l_{ift},$$

where y_{ift} denotes output, l_{ift} labor hired, and φ_f (time-invariant) firm productivity. Importantly, φ_f differs across firms, which captures the notion that some firms are systematically more efficient at producing than others in all of their product lines, e.g., due to a better business plan. As in Aghion et al. (2023), I assume two productivity types, i.e., $\varphi_f \in \{\varphi^h, \varphi^l\}$ where $\varphi^h/\varphi^l > 1$, which I refer to as high- and low-type firms.

4.3 Static allocation

Taking the joint distribution of product qualities and firm productivity as exogenous in this section, I characterize the static allocations at the product, firm and aggregate levels.

4.3.1 Product level

Firms in product market i compete in prices (Bertrand competition). In equilibrium, only the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$ produces product i (henceforth, incumbent). Under Bertrand competition, the incumbent firm engages in limit pricing and sets its price equal to the quality-adjusted marginal costs of the follower (the firm with the second highest quality-adjusted productivity)

$$p_{if} = \frac{q_{if}}{q_{if'}} \frac{w}{\varphi_{f'}}, \tag{3}$$

where f' indexes the follower in product market i. The price that the incumbent sets is increasing in the quality gap between the incumbent and the follower, as eq. (3) shows. Defining the product markup as the output price over marginal costs, it follows

$$\mu_{if} \equiv \frac{p_{if}}{w/\varphi_f} = \frac{q_{if}}{q_{if'}} \frac{\varphi_f}{\varphi_{f'}}. \tag{4}$$

The incumbent's markup for product i is increasing in its quality and productivity gap. The price setting of the incumbent gives rise to the following profits for product i

$$\pi_{if} = p_{if} y_{if} - w l_{if} = Y \left(1 - \frac{1}{\mu_{if}}\right),$$

with labor demand for product i

$$l_{if} = \frac{Y}{w} \mu_{if}^{-1}. \tag{5}$$

Employment in product line i is decreasing in the markup.

4.3.2 Firm level

Firm employment is the sum of employment across the firm's product lines

$$l_f = \sum_{i \in N_f} l_{if} = \frac{Y}{w} \left(\sum_{i \in N_f} \mu_{if}^{-1} \right),$$

where N_f denotes the set of product lines in which firm f is the incumbent producer. Firm employment decreases in the markups within each product line but increases in the number of product lines. Hence, holding product markups constant, firms that produce in more product lines feature higher employment. Vice versa, holding the number of product lines constant, firms with higher product markups employ less labor. As sales are equalized across product lines, firm sales are given by $\left|N_f\right| Y \equiv n_f Y$, where n_f denotes the number of products firm f is producing. Hence, firms that produce in more product lines feature higher sales. Lastly, I define the firm markup μ_f as total firm sales over the wage bill wl_f.

4.3.3 Aggregate level

Integrating employment across firms or products yields the total workforce in production:

$$L_P = \int_f l_f df = \frac{Y}{w} \int_0^1 \mu_{if}^{-1} di. \tag{6}$$

Taking logs and integrating eq. (4), one obtains an expression for the wage

$$w = \exp\left(\int_0^1 \ln q_{if} di\right) \times \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right) \times \exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right). \tag{7}$$

To find an expression for aggregate output, insert eq. (7) into eq. (6) to obtain

$$Y = Q\Phi\mathcal{M}L_P, \tag{8}$$

where

$$Q = \exp\left(\int_0^1 \ln q_i di\right), \quad \Phi = \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right), \quad \mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_i^{-1} di\right)}{\int_0^1 \mu_i^{-1} di}.$$

Aggregate output Y depends on geometric averages of quality Q and productivity Φ as well as on misallocation $\mathcal{M}$ and production labor L_P. Misallocation arises from markup dispersion ($\mathcal{M}$ is bounded by unity from above) that is due to quality and productivity heterogeneity. The product of Q, Φ and $\mathcal{M}$ captures aggregate Total Factor Productivity (TFP).

4.4 Dynamic firm problem

Firms compete for product markets through innovation (R&D). Via expansion R&D, a firm improves the quality of a randomly selected product operated by a competitor, thereby becoming the highest-quality producer in a product line that is new to the firm. Following Peters (2020), firms also undertake internal R&D, which enables them to further raise the quality of products they already control after successful expansion. Internal R&D increases product markups according to eq. (4), creating a wedge between sales and wage-bill growth. This mechanism allows the model to match the distinct sales (value added) and wage bill trajectories—and their evolution over time—shown in Figure 1.

Item quality is improved step-wise such that every innovation (external or internal) increases q_i by a factor of λ.[8] Denoting by λ^{Δ_i} the relative qualities of incumbent and second-best

[8] As Aghion et al. (2023), I assume that the step size of quality improvements exceeds the productivity differential, $\lambda > \varphi^h/\varphi^l$. This assumption ensures that the firm with the highest quality version in a product line is the incumbent producer. Relaxing this assumption would give room for a race for incumbency between low-productivity entrants facing a high-productivity incumbent from which I abstract.

firms within a product line, i.e.,

$$\lambda^{\Delta_i} = \frac{q_{if}}{q_{if\prime}}$$

and by $[\mu_i]$ a firm's set of markups across its product lines, firm profits follow

$$\pi_f(n, [\mu_i]) \equiv \sum_{k=1}^{n} \pi_i(\mu_k) = \sum_{k=1}^{n} Y\left(1 - \frac{1}{\mu_k}\right) = \sum_{k=1}^{n} Y\left(1 - \frac{1}{\lambda^{\Delta_k}} \frac{1}{\frac{\varphi_{f(k)}}{\varphi_{f'(k)}}}\right).$$

Incumbent firms choose the rate of expansion R&D, x_{it}, and the rate of internal R&D, I_{it}, for each of their product lines, i. When choosing x_{it} and I_{it}, firms take aggregate output Y_t, the real wage w_t, the share of lines operated by high-productivity firms S_t, the interest rate r_t and the rate of creative destruction τ_t as given. Note that S_t is predetermined in t and its evolution is characterized shortly. Denoting the time derivative by $\dot{V}_t^h()$, the value function of a high-productivity type firm (indexed by h) satisfies the following HJB equation:

$$\begin{aligned}
&r_t V_t^h(n, [\mu_i], S_t) - \dot{V}_t^h(n, [\mu_i], S_t) = \\
&\quad \sum_{k=1}^{n} \underbrace{\pi_i(\mu_k)}_{\text{Flow profits}} + \sum_{k=1}^{n} \tau_t \underbrace{\Big[V_t^h\left(n-1, [\mu_i]_{i\neq k}, S_t\right) - V_t^h(n, [\mu_i], S_t)\Big]}_{\text{Creative destruction}} \\
&\quad + \max_{[I_{it}, x_{it}]} \Bigg\{ \sum_{k=1}^{n} I_{kt} \underbrace{\Big[V_t^h\left(n, [[\mu_i]_{i\neq k}, \mu_k \cdot \lambda], S_t\right) - V_t^h(n, [\mu_i], S_t)\Big]}_{\text{Internal R\&D}} \\
&\quad + \sum_{k=1}^{n} x_{kt} \underbrace{\Bigg[S_t V_t^h\left(n+1, [[\mu_i], \lambda], S_t\right) + (1-S_t) V_t^h\left(n+1, \left[[\mu_i], \lambda \cdot \frac{\varphi^h}{\varphi^l}\right], S_t\right) - V_t^h(n, [\mu_i], S_t)\Bigg]}_{\text{Expansion R\&D}} \\
&\quad - \underbrace{w_t \Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right)}_{\text{R\&D costs}} \Bigg\}.
\end{aligned}$$

The value of a firm consists of flow profits, research costs, and three parts related to internal R&D, expansion R&D, and creative destruction. At the rate of creative destruction τ_t (determined in equilibrium), the firm loses one of its n products, in which case, it remains with $n-1$ products. At the optimally chosen rate I_{it}, internal R&D turns out successful (third row), and the firm charges a λ times higher markup on its product according to eq. (4). Alternatively, at the optimally chosen rate x_{it}, expansion R&D is successful (fourth row), and the firm acquires a new product (n increases by one).

Productivity-type heterogeneity introduces novel elements to the value function of the firm. The share of product lines operated by each productivity type S_t is a state variable that firms keep track of to build markup expectations. When taking over a new product line through expansion R&D (fourth row), the probability of replacing a high-type incumbent is S_t, in which case the high-type entrant charges a markup of λ. With probability $1 - S_t$, the replaced incumbent is of the low type, and the high-type entrant charges a markup of $\lambda \cdot \varphi^h/\varphi^l$. Firms take S_t as given; however, they affect it through their expansion R&D efforts x_{it} in equilibrium. The HJB equation for a low-productivity firm follows the same structure and is listed in the Appendix, Section C.1. The term related to expansion R&D (fourth row) is distinct since markup expectations differ for low-productivity firms.

$\Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right)$ denote the R&D costs. For their R&D activities, firms pay a cost of

$$\Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right) = \sum_{k=1}^{n} c\left(x_{kt}, I_{kt}; \mu_k\right) = \sum_{k=1}^{n} \left[\mu_k^{-1} \frac{1}{\psi_I}(I_{kt})^{\zeta} + \frac{1}{\psi_x}(x_{kt})^{\zeta}\right]$$

in terms of labor. $\zeta > 1$ ensures convexity of the cost function. R&D costs are additively separable to render a closed-form solution of the value function along the balanced growth path. ψ_I and ψ_x scale internal and external R&D costs and capture the R&D efficiency.[9]

The rate of creative destruction τ_t that firms take as given in their optimization problem is, in equilibrium, equal to the sum of expansion R&D rates and the rate of entry z_t

$$\tau_t = \int_0^1 x_{it} di + z_t. \tag{9}$$

Firm entry is determined as follows. Potential entrants produce a flow rate of entry z_t using a technology that is linear in labor: $z_t = \psi_z L_{Et}$, where ψ_z denotes the entry efficiency and L_{Et} research labor of entrants. Entrants improve the quality of a randomly selected product line. The productivity type is realized after entry and assigned with the exogenous probabilities p^h and $1-p^h$, respectively. Entrants start with a one-step quality gap. When $z_t > 0$, the free entry condition requires that the expected value of firm entry equals the entry costs

$$p^h E[V_t^h(1, \mu_i)] + (1 - p^h) E[V_t^l(1, \mu_i)] = \frac{1}{\psi_z} w_t, \tag{10}$$

where the expected value of entering as a high- or low-type firm is

$$\begin{aligned} E[V_t^h(1, \mu_i)] =& S_t V_t^h(1, \lambda) + (1 - S_t) V_t^h(1, \lambda \times \varphi^h/\varphi^l) \\ E[V_t^l(1, \mu_i)] =& S_t V_t^l(1, \lambda \times \varphi^l/\varphi^h) + (1 - S_t) V_t^l(1, \lambda). \end{aligned}$$

Firms' expansion and internal R&D efforts generate a two-dimensional distribution of quality

[9]Since product-line profits are concave in the markup, internal R&D incentives decline with successful innovations. This would imply slowing markup growth at older ages and faster wage-bill growth relative to sales—patterns not supported in Figure 1. I therefore follow Peters (2020) and scale internal R&D costs by the inverse markup, which removes the dependence of internal R&D on product-line markups while preserving its sensitivity to all other equilibrium forces.

and productivity gaps across product lines, denoted by $\nu_t(\Delta, \frac{\varphi_f}{\varphi_{f'}})$. These gaps determine product markups, so the distribution ν_t characterizes the overall markup distribution in the economy. I derive ν_t as a function of firm policies in Appendix C.2. Importantly, the marginal distributions of ν_t determine S_t, the share of product lines operated by high-productivity firms, which firms take as given in their optimization problem. The law-of-motion for S_t follows from ν_t:

$$\dot{S}_t = \sum_{i=1}^{\infty} \left[\dot{\nu}_t \left(i, \frac{\varphi^h}{\varphi^h} \right) + \dot{\nu}_t \left(i, \frac{\varphi^h}{\varphi^l} \right) \right] = S_t x_t^h + z_t p^h - S_t \tau_t. \tag{11}$$

Here, S_t increases through the expansion of high-type incumbents into new product lines and the entry of new high-type firms, and decreases due to creative destruction.

Lastly, labor market clearing requires that production labor L_{Pt} and total research labor L_{Rt} add up to one, the aggregate labor endowment

$$1 = L_{Pt} + L_{Rt} = \frac{Y_t}{w_t} \int_0^1 \mu_i^{-1} di + \int_0^1 \left(\mu_i^{-1} \frac{I_{it}^{\zeta}}{\psi_I} + \frac{x_{it}^{\zeta}}{\psi_x} \right) di + \frac{z_t}{\psi_z}. \tag{12}$$

4.5 Balanced growth path characterization

I define a balanced growth path of the economy as follows.

Definition 1. *A balanced growth path (BGP) is a set of allocations $[x_{it}, I_{it}, \ell_{it}, z_t, S_t, y_{it}, C_t]_{it}$ and prices $[r_t, w_t, p_{it}]_{it}$ such that firms choose $[x_{it}, I_{it}, p_{it}]$ optimally, the representative household maximizes utility choosing $[C_t, y_{it}]_{it}$, the growth rate of aggregate variables is constant, the free-entry condition holds, all markets clear and the distribution of quality and productivity gaps ν_t is stationary.*

Along the balanced growth path, the economy can be characterized in closed form.

Proposition 1. *In the above setup, along a balanced growth path:*

1. *The value of a product line for a firm of productivity type $d \in \{h, l\}$ is given by*

$$V_i^d(1, \mu_i, S) = \frac{1}{\rho + \tau} \left[Y_t \left(1 - \frac{1}{\mu_i} \right) + \frac{\zeta - 1}{\psi_x} x_i^{\zeta} w_t + \frac{\zeta - 1}{\psi_I} I_i^{\zeta} w_t \mu_i^{-1} \right] \tag{13}$$

   The value of a firm is the sum of its product line values $V^d(n, [\mu_i], S) = \sum_{k=1}^{n} V_i^d(1, \mu_k, S)$.

2. *Expansion R&D rates are productivity-type specific, i.e., $x_i \in \{x^h, x^l\}$. Further, high-productivity firms expand faster than low-productivity ones $x^h > x^l$.*

3. *The share of product lines operated by high-productivity type firms S is determined by*

$$S\tau = Sx^h + zp^h. \tag{14}$$

4. The growth rate of aggregate variables is given by

$$g = \frac{\dot{Q}_t}{Q_t} = \left(\underbrace{I}_{\text{Incumbent internal R\&D}} + \underbrace{Sx^h + (1-S)x^l}_{\text{Incumbent expansion R\&D}} + \underbrace{z}_{\text{Entry}} \right) \times \ln(\lambda). \tag{15}$$

Proof. The Appendix, Sections C.1, C.2 and C.3 contain the proofs. □

The value of a product line in eq. (13) consists of three terms: profits for a given markup and the continuation values of expansion and internal R&D. The sum of the three terms is discounted by the discount rate and the rate of creative destruction. The more impatient the household or the higher the risk of replacement, the lower the value of a product line. Importantly, expansion R&D rates are productivity-type specific. More productive incumbents charge higher markups, which increases the expected value of a product line. The optimality condition for expansion R&D (Appendix, eq. A-6) equates the expected value of a product line with the marginal cost of expansion R&D. Hence, in equilibrium, more productive firms pay a higher marginal cost of expansion R&D and choose $x^h > x^l$. By contrast, internal R&D rates are type-invariant, $I_i \equiv I$. Because expansion rates differ by type, the values of both a product line and a firm are type specific as well.

Proposition 1 further shows that the share of product lines operated by high-productivity type firms is constant along the balanced growth path. The expression in eq. (14) directly follows by setting the law-of-motion for S_t in eq. (11) equal to zero. Eq. (14) captures that, along the balanced growth path, the share of product lines by high-productivity firms that fall victim to creative destruction is exactly equal to the share of product lines that high-productivity firms gain by entering new product lines through incumbent creative destruction or firm entry. Eq. (14) can further be rearranged to

$$S = \frac{zp^h}{(1-S)(x^l - x^h) + z}, \tag{16}$$

which shows that S depends on the difference in the expansion R&D rates between firm types, $x^l - x^h$. Holding firm entry z fixed, an increase in the expansion rate of high-productivity incumbents must be matched by an equal rise in the expansion rate of less-productive firms for S to remain constant. Note that positive firm entry is necessary for both firm types to co-exist in equilibrium where $x^l \neq x^h$ and S is constant at its interior solution.

Long-run growth results from R&D at the product level. This occurs through successful internal R&D or (incumbents' and entrants') creative destruction. The aggregate arrival rate of innovation is, hence, equal to the sum of the rates of creative destruction τ and internal R&D I. Multiplying the arrival rate by the log step size of innovation delivers the aggregate growth rate g, as shown in eq. (15) of Proposition 1. Since expansion R&D rates are heterogeneous, i.e., $x^h > x^l$, changes in the share of product lines operated by each productivity type, S and $1-S$, affect the aggregate growth rate. Along the balanced growth path, both τ and g are constant. I further characterize the aggregate labor income share, the TFP misallocation measure $\mathcal{M}$, and the aggregate markup analytically in the Appendix, Section C.2.

To find the balanced growth path, I jointly solve the optimality conditions of the firm (derived in Appendix C.1), the free entry condition, eq. (10), the labor market clearing condition, eq. (12), and the system of differential equations characterizing the distribution of productivity and quality gaps captured in eqs. (A-8) and (A-9). Appendix C.4 contains the details.

4.6 Firm dynamics

Firms lose products according to the same stochastic process as in Klette and Kortum (2004). However, in this model, firms add products at systematically different rates as optimally chosen expansion R&D rates vary with the firm's productivity type.[10] The following section derives productivity-type specific expressions for firm growth and survival probabilities.

4.6.1 Firm sales growth

Firm sales are proportional to the number of products a firm produces. As such, successful expansion R&D increases firm sales. Since optimal expansion R&D rates are productivity-type specific, so is sales growth. Conditional on survival, expected sales growth for a firm with productivity φ_f between age zero and age a_f is

$$E\left[\ln n_f Y | a_f, \varphi_f\right] - E\left[\ln n_f Y | 0, \varphi_f\right] = \underbrace{g \times a_f}_{\text{Aggregate growth}} + \underbrace{E\left[\ln n_f | a_f, \varphi_f\right]}_{\text{Firm's product growth}},$$

where n_f is the firm's number of products. The probability of producing n products at age a conditional on survival is $\left(1-\gamma^j(a)\right)\left(\gamma^j(a)\right)^{n-1}$, where $\gamma^j(a) = x^j \frac{1-e^{-(\tau - x^j)a}}{\tau - x^j e^{-(\tau - x^j)a}}$ and $j \in \{h, l\}$. Therefore, expected sales growth conditional on survival for productivity type φ_f is

$$E\left[\ln n_f Y | a_f, \varphi_f\right] - E\left[\ln n_f Y | 0, \varphi_f\right] = \underbrace{g \times a_f}_{\text{Aggregate growth}} + \underbrace{\left(1-\gamma^f(a_f)\right)\sum_{n=1}^{\infty} \ln n \times \left(\gamma^f(a_f)\right)^{n-1}}_{\text{Firm's product growth}}. \tag{17}$$

[10]Therefore, the properties related to firm-size growth and survival in Klette and Kortum (2004) hold conditional on the firm type. In particular, conditional on the type, firm size, and growth are unrelated, as in Lentz and Mortensen (2008). For the unconditional firm-size and growth correlation, two forces are at play. On the one hand, young (small) firms tend to grow quicker due to the survival bias. On the other hand, more productive firms (with faster growth rates) are more likely to end up large. In the estimated model, the majority of the firms are of the high-productivity type. Hence, size is unrelated to growth for most firms.

4.6.2 Firm employment growth

Average employment conditional on age and productivity type is equal to

$$E[\ln l_f|a_f,\varphi_f] = \ln\left(\frac{Y}{w}\right) + E\left[\ln n_f|a_f,\varphi_f\right] - E\left[\ln \mu_f|a_f,\varphi_f\right].$$

Since $\frac{Y}{w}$ is constant, employment growth simplifies to

$$E[\ln l_f|a_f,\varphi_f] - E[\ln l_f|0,\varphi_f] = \underbrace{E\left[\ln n_f|a_f,\varphi_f\right]}_{\text{Firm's product growth}} - \underbrace{\left(E\left[\ln \mu_f|a_f,\varphi_f\right] - E\left[\ln \mu_f|0,\varphi_f\right]\right)}_{\text{Firm's markup growth}}, \quad (18)$$

where $E\left[\ln n_f|a_f,\varphi_f\right]$ is defined in eq. (17) and firm markup growth, $E\left[\ln \mu_f|a_f,\varphi_f\right] - E\left[\ln \mu_f|0,\varphi_f\right]$, is derived in Appendix C.5. As wages and output grow at the same rate along the balanced growth path, eqs. (17) and (18) further imply that growth in the firm's wage bill is equal to sales growth minus markup growth.

4.6.3 Firm survival

Firms that lose their last product exit the economy. Since firm expansion is type-dependent, so is firm survival. The survival function in Klette and Kortum (2004) holds conditional on the firm type, i.e., the share of high and low type firms surviving until age a_f is

$$\chi^h(a_f) = 1 - \tau\frac{1-e^{-(\tau-x^h)a_f}}{\tau - x^h e^{-(\tau-x^h)a_f}} \quad \text{and} \quad \chi^l(a_f) = 1 - \tau\frac{1-e^{-(\tau-x^l)a_f}}{\tau - x^l e^{-(\tau-x^l)a_f}}. \quad (19)$$

Hence, the share of high-type firms among surviving firms at age a_f is

$$s^h(a_f) = \frac{p^h\chi^h(a_f)}{p^h\chi^h(a_f) + (1-p^h)\chi^l(a_f)}, \quad (20)$$

which corresponds to the mass of high-type survivors relative to the total mass of survivors. Since $x^h > x^l$, it is easy to show that sales growth conditional on survival of high-productivity firms exceeds that of low-productivity ones and that the share of the former among surviving firms $s^h(a_f)$ increases in a_f.

4.6.4 Average firm-size trajectory

Firm-size growth and survival probabilities by productivity type characterize the average firm-size trajectory. Average firm size conditional on age relative to the size at entry is

$$E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t} = a_f\right] - E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t} = 0\right] = \quad (21)$$

$$
\begin{aligned}
& s^h(a_f)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^h\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right]\right)}_{\text{Size growth conditional on survival (high-productivity type)}} \\
& +\left(1-s^h(a_f)\right)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=a_f,\varphi_f=\varphi^\ell\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Size growth conditional on survival (low-productivity type)}} \\
& +\left(s^h(a_f)-s^h(0)\right)\times \underbrace{\left(E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^h\right]-E\left[\ln \text{Size}_{f,t}|\text{Age}_{f,t}=0,\varphi_f=\varphi^\ell\right]\right)}_{\text{Correction term}}.
\end{aligned}
$$

Firm size relative to size at entry is equal to the sum of firm growth conditional on survival of high- and low-productivity firms over the first a_f years, weighted by their share among surviving firms, $s^h(a_f)$ and $1-s^h(a_f)$ and a correction term. This last term corrects for size differences at entry between both productivity types. If firm size is measured using sales, the last term disappears since entrants of both productivity types start with equal sales. Hence, eq. (21) highlights that the average firm size trajectory steepens if size growth conditional on survival of either productivity type increases or $s^h(a_f)$, the share of high-productivity firms among survivors, rises since their size growth conditional on survival exceeds that of low-productivity firms.

4.6.5 Firm size distribution

The theory makes precise predictions about the firm-size distribution. I derive the firm-size distribution in Section C.7 in the Appendix. Denoting by M^h the mass of high-productivity type firms and by M the total mass of firms, the share of high-productivity type firms is

$$S_{M^h}=\frac{M^h}{M}, \tag{22}$$

and the firm entry rate

$$\text{Firm entry rate}=\frac{z}{M}. \tag{23}$$

5 Balanced growth path application

This section applies the model to explain the documented changes in the average firm-size trajectory and the macroeconomy, particularly rising firm-size dispersion, quantitatively. To this extent, I estimate the model along two balanced growth paths. The initial balanced growth path captures firm-size dynamics and macroeconomic conditions during the 1990s. I then re-estimate model parameters to reflect the changes in firm-size dynamics and macroeconomic conditions with respect to the 2010s.

5.1 Initial balanced growth path

There are, in total, eight parameters in the model. The internal R&D efficiency ψ_I, the expansion R&D efficiency ψ_x, the innovation cost curvature ζ, the entry efficiency ψ_z, the step size of quality improvements λ, the productivity differential φ^h/φ^ℓ, the share of high-productivity type firms among entrants p^h, and the discount rate ρ. Three of these parameters are calibrated exogenously. First, the discount rate ρ is set to 0.05. Second, I follow Acemoglu et al. (2018) who set ζ equal to two based on evidence from the microeconometric innovation literature (Blundell et al., 2002; Hall and Ziedonis, 2001). Third, Figure 3 shows that firm size at entry has remained very stable over time, indicating no systematic changes in the nature of entrants. Therefore, I fix the share of high-productivity firms among entrants p^h at some exogenous level. Since the interquartile range as the measure of firm-size dispersion captures dispersion around the median firm, I set p^h to 0.5. Hence, one can interpret high- and low-productivity firms as above or below median firms at entry. The results in this paper are similar for alternative values of p^h.

The other five parameters are estimated, targeting the average firm-size trajectory (value added and the wage bill), cross-sectional firm-size dispersion, the firm entry rate, and aggregate TFP growth. Even though all parameters are identified jointly, there is a tight mapping between parameters and targets.

Matching trajectories of value added and the wage bill disciplines the firms' R&D efficiencies ψ_x and ψ_I. In the model, successful expansion R&D translates into firm sales growth. In the estimation, ψ_x adjusts expansion R&D costs such that sales trajectories match a given target. Since labor is the only production input in the model, firm sales equal value added, and I use the latter as the measure in the data. Moreover, the internal R&D costs govern firms' markup growth. Since the firm markup is the wedge between value added (sales) and the wage bill in the model, targeting growth in value added and the wage bill jointly disciplines firm markup growth and, hence, the internal R&D efficiency ψ_I. The advantage of targeting the wage bill instead of markups is that the former is directly observed in the data. I target the average value added and wage bill of firms aged eight relative to entry for the cohorts 1997 to 2001. According to Figure 1, value added increased by 0.492 and the wage bill by 0.468 log points over the first eight years. Eight years is long enough to reflect firm size growth and still allow for estimating separate balanced growth paths (one for the early cohorts and one for the latest cohorts) over the data coverage period from 1997 to 2017. The model matches the firm-size dynamics over the entire life cycle rather well, so the specific age targeted is not consequential.

To pin down the productivity differential φ^h/φ^ℓ, I target the firm-size dispersion observed in the data. To see the link, note that the targeted firm-size trajectories generate cross-sectional size dispersion. To the extent that the observed size dispersion in the data exceeds the model-generated value for the targeted size-trajectories, the model asks for an increase in the variance of firm size conditional on age by increasing the productivity differential. As the measure of size dispersion, I target the interquartile range of log value added within two-digit industries. Size dispersion in the data is enormous. To make the size dispersion in the data comparable to the model, I normalize a firm's value added by its respective value added at entry. This normalization equalizes firm size at entry in the data and resembles the fact that, as common in Klette and Kortum (2004) style models, all firms start with one product. Without this normalization, cross-sectional size heterogeneity in the data that is due to size dispersion at firm entry is loaded on the productivity differential φ^h/φ^ℓ. Since the year of entry is left-censored in the data, I take the average within-industry interquartile range over the most recent years (2010–2017) and subtract the observed increase in Figure 4b. This results in an interquartile range of 0.742, i.e., the 75th-percentile firm is roughly $e^{0.742} \approx 2.1$ times as large as the 25th-percentile firm in 1997. In the model, the interquartile range is restricted to integer values.[11] Since the value in the data is very close to log(2), I target this value for the initial balanced growth path.[12]

Table 2: Initial balanced growth path. Moments and parameters

	Data	Model
Moments		
Value added at age 8 relative to entry in logs	0.492	0.492
Wage bill at age 8 relative to entry in logs	0.468	0.468
Interquartile range of log value added	log(2)	log(2)
Entry rate in %	14.8	14.8
TFP growth g in %	1.5	1.5
Parameters		
ψ_x *Expansion R&D efficiency*		0.892
ψ_I *Internal R&D efficiency*		0.925
φ^h/φ^ℓ *Productivity gap*		1.019
ψ_z *Entry R&D efficiency*		3.790
λ *Step size of quality improvements*		1.051
Set exogenously		
ρ *Discount rate*		0.05
ζ *R&D cost curvature*		2
p^h *Share of high-type firms among entrants*		0.5

Notes: the targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

Identification of the remaining two parameters is straightforward. The entry efficiency ψ_z

[11] For example, if the firm at the 25th percentile of the cross-sectional value-added distribution produces one product, the interquartile range of log value added is equal to log(2) if the 75th percentile firm produces two products, log(3) if the 75th percentile firm produces three products and so forth.

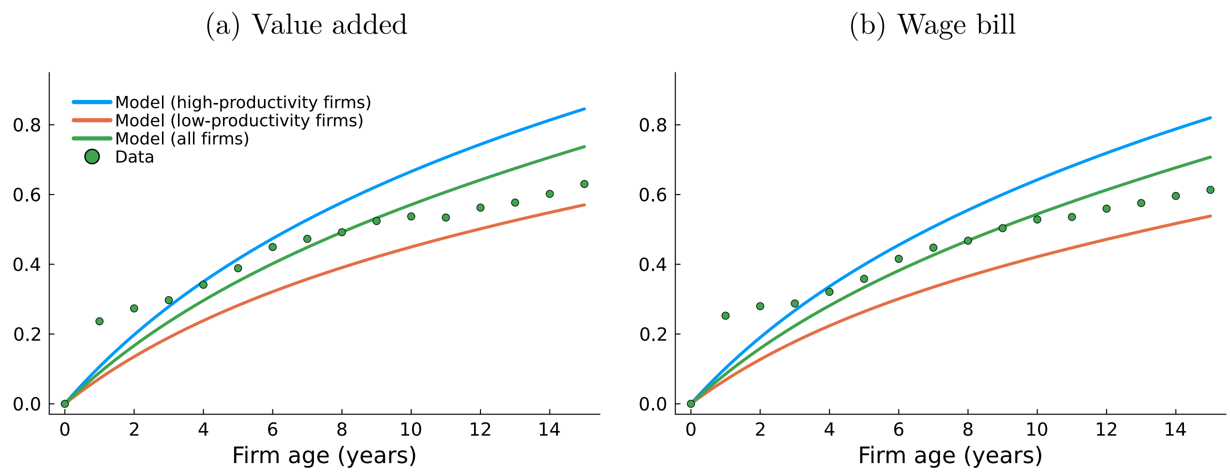
[12] To improve the computational efficiency of the estimation, I target a value for the standard deviation of log value added instead of the interquartile range and check afterward that the latter equals log(2) along the balanced growth path. The advantage of the former is that it provides finer gradients of the objective function in the parameter search. Computing the empirical counterpart of the standard deviation would require dropping firms with negative value added.

determines the cost of a given flow rate of entry in units of labor. I target an entry rate of 14.8%, the value in 1997 in Figure 5a, to pin down ψ_z. Lastly, aggregate TFP growth disciplines the step size of quality improvements λ: the aggregate growth rate g in eq. (15) directly depends on λ. I target an aggregate growth rate of 1.5%, the average over the period 1995–2005 in Figure 5b. Note that in this estimation, the average firm-size trajectory is targeted directly, the firm-age distribution is disciplined by the targeted entry rate, and size heterogeneity conditional on age is determined residually by targeting overall size dispersion. All targets are summarized in Table 2.

The estimation follows a two-step approach. In the first (global) step, the algorithm computes the sum of squared percentage deviations from the targeted moments for a large Sobol sequence of parameter vectors. All targets receive equal weights. In the second (local) step, I take the best candidates from the first step and perform a local search. The local search, again, minimizes the distance from the targets. The estimation is robust, i.e., the best parameter vectors from the second step converge to the same parameter values.

Table 2 shows the estimation results. The model replicates all targeted moments exactly. Two parameters can be interpreted intuitively: successful innovation increases product quality by 5.1%, and the productivity differential φ^h/φ^ℓ equals roughly 2%. The productivity advantage is relatively small, which relates to the fact that I target firm-size dispersion in the data that is not accounted for by size dispersion at entry.[13] Nevertheless, firms of the two types differ significantly over the life cycle, as shown below.

Figure 6: Firm size relative to entry



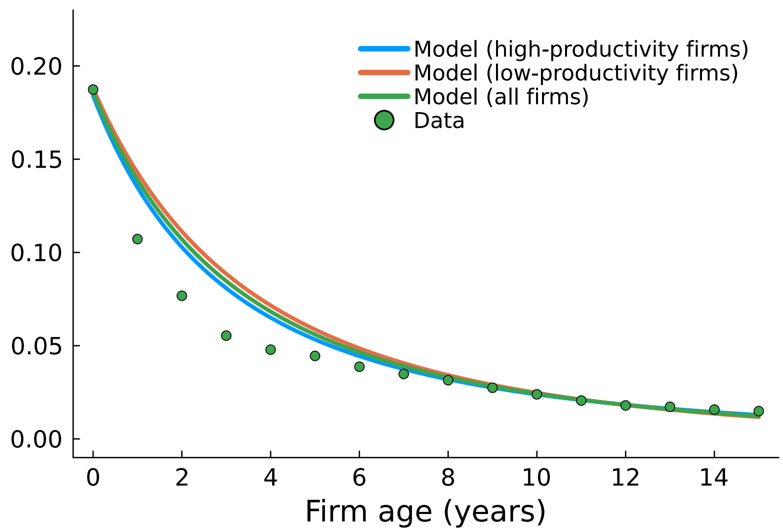
Notes: the figure shows the average value added (left) and wage bill (right) relative to entry (in logs) as a function of firm age in the model (initial balanced growth path) and the data (cohorts 1997–2001 in Swedish registry data). The size at age eight relative to entry has been targeted.

Along the balanced growth path, high-productivity firms constitute 56% of all firms, S_{M^h} in eq. (22), and capture 65% of total value added, as measured by S. These numbers are

[13] Another reason why the productivity gap is relatively small compared to, e.g., Aghion et al. (2023) is that their model abstracts from firm markup growth, so the productivity differential plays a larger role in explaining the aggregate markup, which is targeted in their estimation.

larger than their share at entry ($p^h = 0.5$) due to high-productivity firms expanding faster into new product lines than less productive ones: $x^h - x^\ell = 0.053$. This is reflected in their size growth. Conditional on survival, value added grows on average by 0.58 log points for high-type firms but only by 0.39 log points for low-type firms over the first eight years. Figure 6 shows the trajectories of value added and the wage bill for each productivity type over the entire life cycle. The figure clearly illustrates the heterogeneity in growth profiles. Conditional on survival, the difference in value added growth over the first 15 years between both productivity types accumulates to roughly 0.3 log points. Figure 6 includes the size trajectories in the data (cohorts 1997–2001). The model slightly misses the concavity of the trajectories for the first two years; however, the fit over the entire life cycle is overall good despite being untargeted (except for age eight).

Figure 7: Exit rates by firm age



Notes: the figure shows the firm exit rates as a function of age in the estimated model (initial balanced growth path) and the data (cohorts 1997–2001). The exit rates are computed as the increments in a Kaplan-Meier survival function.

The model also performs well along other untargeted dimensions. Figure 7 shows the firm exit rates by age in the model and the data (cohorts 1997-2001). The model slightly overshoots the exit rates for the first few years, consistent with undershooting firm size growth in Figure 6 for these years. However, the overall fit over the entire life cycle is good. In the model, faster size growth by the more productive firms translates into lower exit rates, as Figure 7 shows. The difference in exit rates between both productivity types accumulates over time. The share of high-productivity firms that survive until age 15 is about five percentage points higher than the share of low-productivity ones.

Next, I conduct a set of comparative-statics exercises around the initial balanced growth path to illustrate how different model parameters affect firm-size dispersion. I vary each parameter individually by 20%, choosing the direction of adjustment such that firm-size dispersion increases relative to the initial balanced growth path. The left panel of Table 3 reports selected equilibrium outcomes. As my summary measure of firm-size dispersion, I use the standard deviation of log value added across firms, denoted σ_{VA}. The right panel of Table 3 decomposes the resulting change in σ_{VA} into its components following eq. (1). Each entry reports the share (in percent) of the total change in dispersion (relative to the initial balanced growth path) attributable to changes in the average firm-size trajectory, $E[\ln s|a]-E[\ln s|0]$, the age distribution, $p(a)$, or within-age firm-size dispersion, $\mathrm{Var}(\ln s|a)$.[14] Because the decomposition is not additive, the shares do not necessarily sum to 100%.

Table 3: Comparative statics in the initial balanced growth path

	x^h	x^ℓ	$\frac{z}{M}$	$s_{a\geq 10}$	σ_{VA}	Decomposing changes in σ_{VA} (in %) $E[\ln s\|a]-E[\ln s\|0]$	$p(a)$	$\mathrm{Var}(\ln s\|a)$
Initial BGP	0.144	0.091	0.148	0.212	0.5			
Parameter Δ								
$\psi_x \uparrow$	0.177	0.105	0.134	0.224	0.63	37.8	21.2	36.9
$\varphi^h/\varphi^\ell \uparrow$	0.15	0.085	0.144	0.217	0.53	23.5	35.2	39.4
$\psi_z \downarrow$	0.182	0.1	0.075	0.295	1.18	12.4	58.8	18.6

Notes: the left table shows balanced growth path (BGP) outcomes (columns) for different model parametrizations (rows). x^h and x^ℓ denote the expansion R&D rates of high- and low-productivity firms, $\frac{z}{M}$ the entry rate, $s_{a\geq 10}$ the share of firms aged ten years or older defined by the cdf $p^h(1-\chi^h(a_f))+(1-p^h)(1-\chi^\ell(a_f))$ and σ_{VA} the standard deviation of log value added across firms. The parameters ψ_x and ψ_z denote the expansion and entry R&D efficiency, respectively. φ^h/φ^ℓ is the productivity differential. The size of the parameter change is 20% (for φ^h/φ^ℓ the change refers to the parameter value minus one). The direction of the change is chosen to generate an increase in σ_{VA} relative to the initial BGP. The right table decomposes the changes in σ_{VA} according to eq. (1). Each number indicates the share (in percent) of the change in σ_{VA} that is due to changes in size trajectories, the age distribution and size dispersion conditional on age, respectively.

First, Table 3 shows that a reduction in expansion costs (an increase in expansion efficiency, ψ_x) raises the expansion rate of high-productivity firms from 0.144 to 0.177 and that of low-productivity firms from 0.091 to 0.105. The firm entry rate, z/M, falls from 14.8% to 13.4%, while the share of firms aged ten or older, $s_{a\geq 10}$, rises from 21.2% to 22.4%. These adjustments are associated with an increase in the standard deviation of log firm size from 0.50 to 0.63. The right panel of Table 3 indicates that the resulting rise in firm-size dispersion is driven primarily by a steepening of the average firm-size trajectory (37.8%). Size dispersion conditional on age also contributes, though to a slightly lesser extent (36.9%). Changes in the firm-age distribution account for the smallest share (21.2%). Taken together, the decomposition indicates that lower expansion costs increase firm-size dispersion primarily by raising both x^h and x^ℓ, which steepens the average firm-size trajectory.

Second, a rise in the productivity gap, φ^h/φ^ℓ, raises the expansion rate of high-productivity firms while lowering that of low-productivity firms. This divergence in expansion rates naturally amplifies size dispersion within age brackets: firms of the same age become more unequal because some expand faster and others more slowly. Accordingly, the largest share of the resulting increase in overall dispersion comes from changes in size dispersion conditional on age (39.4%), whereas movements in the average firm-size trajectory contribute only

[14] For the average firm-size trajectory, to highlight the effects of the displayed changes in firms' expansion rates x^h and x^ℓ, I hold all other equilibrium variables fixed at their initial levels in this section.

modestly (23.5%).

Lastly, an increase in entry costs (a decline in entry efficiency, ψ_z) substantially reduces the firm entry rate to 7.5% and increases the share of firms aged ten or older to 29.5%. Consistent with this pattern, most of the resulting increase in size dispersion is accounted for by the shift in the firm-age distribution (58.8%).

5.2 New balanced growth path

To quantify how much each force contributes to the rise in firm-size dispersion, this section estimates the model along a new balanced growth path that—relative to the previous one—mirrors the documented changes. I target changes in the same five moments used in the initial estimation. The first two targets capture the steepening of the firm-size trajectories (value added and the wage bill) depicted in Figure 1. Specifically, I target the average trajectory of the 2002–2006 and 2007–2011 cohorts. Over the first eight years, value added increases by 0.638 log points and the wage bill by 0.623 log points. These figures exceed their counterparts in the initial balanced growth path, which are 0.492 and 0.468 log points, respectively. As the third moment, I target the change in firm-size dispersion. Figure 4b shows that the interquartile range of value added increased by 0.74 log points. In the model, I match this by raising the interquartile range from log(2) in the initial balanced growth path to log(5) in the new one[15]—an increase of 0.92 log points, which slightly overshoots the observed value. I target log(5) instead of log(4) (which would undershoot the change) to give the firm-age distribution and size dispersion conditional on age more room to explain the increase in total size dispersion, as the firm-size trajectory is targeted separately. Even with this conservative target, the change in the firm-size trajectory turns out to be the main driver. Lastly, I target the fall in the firm entry rate from 15% to 11% shown in Figure 5a and the decline in the aggregate growth rate from 1.5% to 1.1% shown in Figure 5b. I allow all five parameters of the previous estimation to change.

Table 4 captures all targeted moments and parameter estimates of the new balanced growth path. The estimation replicates the changes in all five moments well. The last column of the table compares the estimated parameters with their respective values in the initial balanced growth path. The identified parameter changes are consistent with the literature that studies the drivers behind recent macroeconomic trends. The estimation highlights a fall in the cost of expanding into new product lines (rise in the expansion R&D efficiency) and an increase in the productivity gap between high- and low-type firms. Aghion et al. (2023) arrive at a similar conclusion. Further, Table 4 points to innovations becoming more incremental as the step size of quality improvements declines. This mechanism is emphasized by Olmstead-Rumsey (2019). Bloom et al. (2020) further show that R&D expenditures have increased relative to R&D output, potentially due to innovations becoming more incremental. The estimated increase in the entry R&D efficiency in Table 4 is somewhat surprising at first,

[15] As described for the initial balanced growth path, I target a value for the standard deviation of log value added in the estimation and check afterward that the generated interquartile range equals log(5).

Table 4: New balanced growth path. Moments and parameters

	Data	Model	Initial BGP
Moments			
Value added at age 8 relative to entry in logs	0.638	0.638	0.492
Wage bill at age 8 relative to entry in logs	0.623	0.623	0.468
Interquartile range of log value added	log(5)	log(5)	log(2)
Entry rate in %	11	11	14.8
TFP growth g in %	1.1	1.1	1.5
Parameters			
ψ_x *Expansion R&D efficiency*		1.967	0.892
ψ_I *Internal R&D efficiency*		1.181	0.925
φ^h/φ^ℓ *Productivity gap*		1.021	1.019
ψ_z *Entry R&D efficiency*		6.395	3.790
λ *Step size of quality improvements*		1.036	1.051

Notes: the targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

and I will address it shortly.

To quantify how much each parameter contributes to the changes in the five moments, I implement each estimated parameter change in isolation. Starting from the initial balanced growth path, I measure the contribution of parameter $\theta_i \in \boldsymbol{\theta}$ to moment $M_j(\boldsymbol{\theta})$ as

$$\frac{M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{initial}\right)}{M_j\left(\boldsymbol{\theta}_{-i}^{new}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{initial}\right)} \times 100, \tag{24}$$

where *initial* and *new* index the initial and new balanced growth path.[16] In Appendix E, I start the decomposition from the new balanced growth path and find similar results.

Table 5 reports the contribution of each parameter change (rows) to the five estimation moments (columns), where VA and wL denote the targeted value-added and wage-bill trajectory. Because the model exhibits strong nonlinearities when firm entry becomes low, the contributions do not sum to 100% across rows. As in the comparative-statics exercises above, I measure firm-size dispersion using the standard deviation of log value added across firms, which again allows me to apply the decomposition in eq. (1) in the subsequent analysis.

Table 5 highlights that the rise in the expansion R&D efficiency ψ_x explains the steepen-

[16] Setting ψ_x or λ to their values in the new balanced growth path with the entry R&D efficiency ψ_z kept at its initial balanced growth path value results in zero firm entry. Without entry, the firm-size distribution is non-stationary, i.e., no balanced growth path exists. Hence, for ψ_x and λ, I compute the total contribution by multiplying the contribution of a 0.1% change of the total change in the respective parameter by a factor of 1000. In Appendix E, I perform the decomposition starting from the new balanced growth path instead, which allows one to measure the contributions of ψ_x or λ completely nonlinearly.

Table 5: Decomposition

	VA	wL	σ_{VA}	$\frac{z}{M}$	g
Parameter Δ					
Expansion R&D efficiency $\psi_x \uparrow$	348.4	335.5	107.6	164.1	-37.9
Internal R&D efficiency $\psi_I \uparrow$	2.2	-1.5	-3.4	-19.7	-34.0
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	3.2	2.7	2.9	7.1	1.3
Entry R&D efficiency $\psi_z \uparrow$	-127.2	-116.4	-45.5	-355.7	-97.7
Step size of quality improvements $\lambda \downarrow$	7.9	11.6	33.5	181.3	188.2

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (24). Moments and parameter values of the initial and new BGP are shown in Table 4. VA (wL) denotes the difference in average log value added (wage bill) between firms aged eight and entrants, σ_{VA} the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

ing of the firm-size trajectories (columns VA and wL) and the increase in total firm-size dispersion (column σ_{VA}) jointly. This result is in line with the observation from the previous simple comparative statics that lower expansion costs increase size dispersion mainly by steepening firm-size trajectories. The rise in the expansion R&D efficiency alone accounts for more than the total observed increase in dispersion (107.6%). By contrast, the other parameters—those that operate primarily by increasing size dispersion conditional on firm age (φ^h/φ^ℓ) or through shifts in the firm-age distribution (ψ_z)—contribute only modestly, or even negatively, to the overall increase in dispersion. Consistent with this result, alternatively decomposing the total change in size dispersion into its three endogenous components based on eq. (1)—analogous to the decomposition in Table 3—shows that, across balanced growth paths, 34.4% of the change in σ_{VA} is driven by shifts in the average firm-size trajectory, 28.5% by changes in the age distribution, and 26.2% by movements in dispersion conditional on age.

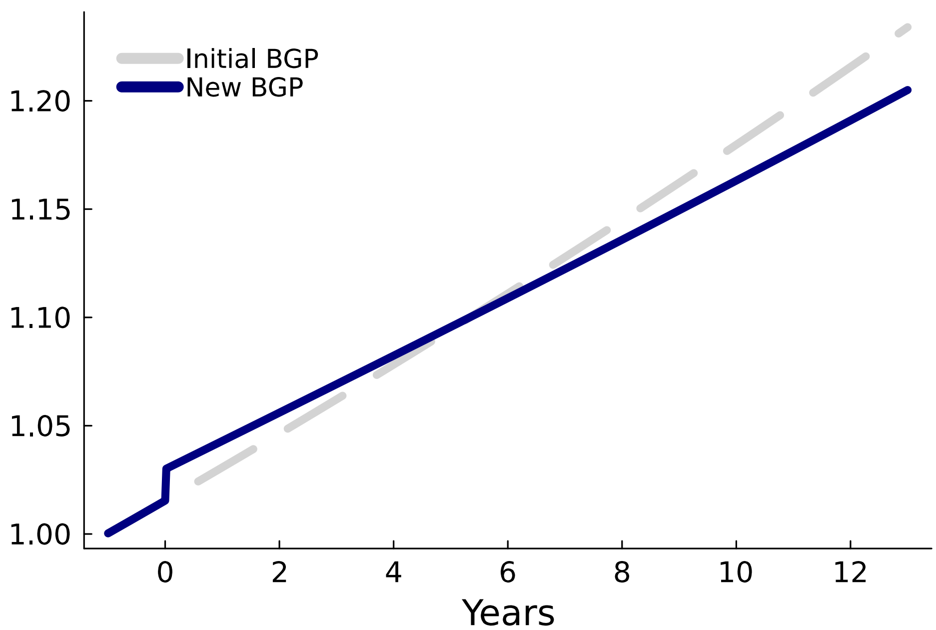
Although the change in expansion R&D efficiency accounts for the rise in firm-size dispersion, it cannot by itself quantitatively match the decline in the firm entry rate (column z/M) or the fall in the aggregate growth rate (column g). On the one hand, higher expansion R&D rates, x^h and x^ℓ, induced by lower expansion costs reduce the equilibrium entry rate z needed to clear the labor market characterized by eq. (12). Quantitatively, this mechanism even lowers entry by more than the observed decline (164.1%). The estimation therefore compensates by reducing entry costs.[17] On the other hand, higher expansion R&D rates raise the aggregate growth rate through equation (15). To offset this effect, the estimation reduces the step size of quality improvements, $\ln \lambda$, which lowers the growth rate one-for-one.

[17] A common interpretation of rising entry costs is that incumbents have erected barriers to entry. An alternative view is that improved access to finance has reduced entry costs for young firms. The model's estimation favors the latter interpretation.

6 Transitional dynamics

The previous section focused on long-run changes in the economy. Although productivity growth declines across balanced growth paths, the fall in expansion costs generates positive growth effects that potentially materialize along the transition. To asses this, I solve for the full transition path to the new balanced growth path. The transition path also provides insights into the welfare implications of moving toward the new balanced growth path.

Figure 8: Transitional dynamics for output

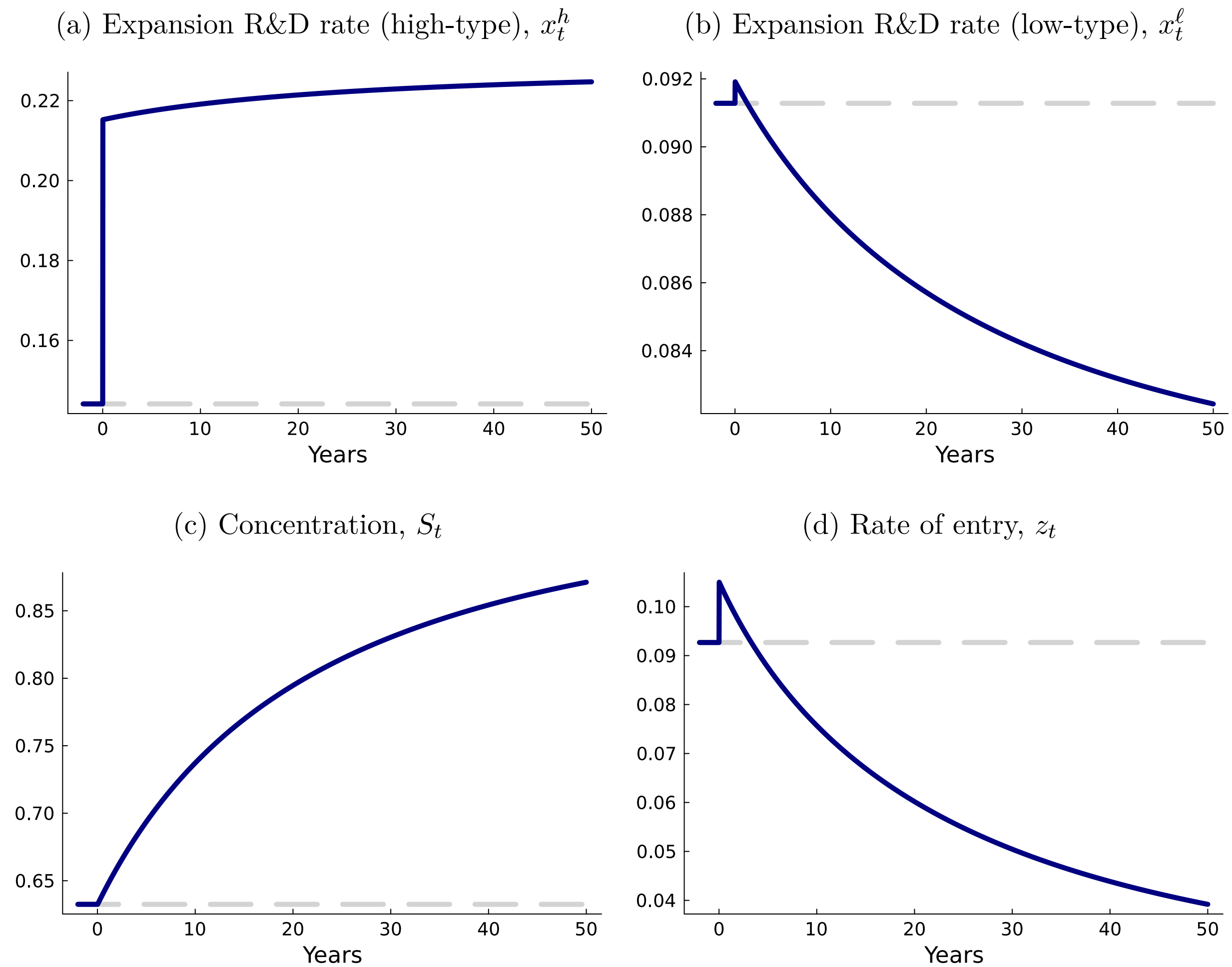


Notes: the figure shows output (consumption) along the transition towards the new balanced growth path. Parameters of the initial and new balanced growth path (BGP) are shown in Table 4. All parameter changes are introduced in period zero. Output is normalized to unity in year minus one for both BGPs.

The algorithm to solve for the transition path works as follows. I solve for firms' policy and value functions from the new balanced growth path backward for a guessed sequence of wage growth, interest rates, and market shares of high-productivity firms, S_t. I then use the obtained policy functions over the transition period to simulate the two-dimensional distribution of quality and productivity gaps forward, starting from the initial balanced growth path. Using the evolution of this distribution over the transition, as well as market clearing and optimality conditions, I back out the implied sequences of wage growth, interest rates, and S_t. The transition path is the fixed point between the guessed and implied sequences. I outline the algorithm in detail in the Appendix, Section D.

Figure 8 shows the transitional dynamics for aggregate output (consumption) during the transition towards the new balanced growth path. Output along the initial balanced growth path is added for comparison in the dashed line. All parameter changes are introduced as one-time shocks in period zero. Indeed, output reacts positively on impact, resulting in a short-run boom in the economy relative to the initial balanced growth path. However, the

Figure 9: Transitional dynamics for equilibrium outcomes



Notes: the figure shows the response in equilibrium outcomes during the transition period towards the new balanced growth path. Parameters of the initial and new balanced growth path are shown in Table 4. All parameter changes are introduced in period zero. The dashed line indicates the initial balanced growth path.

growth burst is short-lived. Already after six years, the output path crosses with the trend of the initial balanced growth path and remains below. Figure 9 provides a closer look at the equilibrium outcomes that drive output growth. As the expansion R&D efficiency increases in period zero, the expansion R&D rates of both high- and low-productivity firms increase on impact. However, the increase is differently pronounced. Expansion R&D rates of high-productivity firms increase substantially, whereas the increase is muted for low-productivity firms. Expansion R&D rates continue to increase over time for high-productivity firms but gradually decrease for less productive ones. The increasing gap in expansion R&D rates results in a reallocation of market shares towards the high-productivity firms. The share of product lines operated by the high type, S_t, gradually increases over time. The bottom right panel shows the flow rate of entry z_t. The increase in the entry R&D efficiency increases z_t on impact; however, the increase is of short duration, as the rise in the expansion R&D rate

x^h puts downward pressure on firm entry through labor market clearing.

That output increases in the short run relative to the initial balanced growth path motivates the question of what the net effect on welfare is. To quantify the change in welfare, I compute the permanent consumption change (in percent) along the initial balanced growth path that makes the representative consumer as well off as with the obtained consumption stream over the transition toward the new balanced growth path. I find that welfare decreases by 4.95%. Hence, the long-run fall in output growth dominates the short-run increase. If one were to compare welfare of the two balanced growth paths without taking the transition into account, the consumption equivalent change ξ is determined by $\ln(1+\xi) = (g^{\text{new}} - g^{\text{initial}})/\rho$, where g^{new} and g^{initial} refer to the long-run aggregate growth rate in each balanced growth path. Given that g declines by 0.4 percentage points across the balanced growth paths and ρ equals 0.05, the welfare loss amounts to 7.69% ($\xi = -0.0769$). Hence, the transition alleviates roughly one-third of the long-run welfare loss.

7 Extensions

7.1 Indirect evidence on falling expansion costs

The previous estimation indicates that declining firm expansion costs are the primary force behind the observed steepening of the average firm-size trajectory and the increase in firm-size dispersion. Although direct evidence on expansion costs is difficult to obtain, I examine an additional endogenous margin that one would expect to respond to lower expansion costs: the number of establishments operated by a firm.

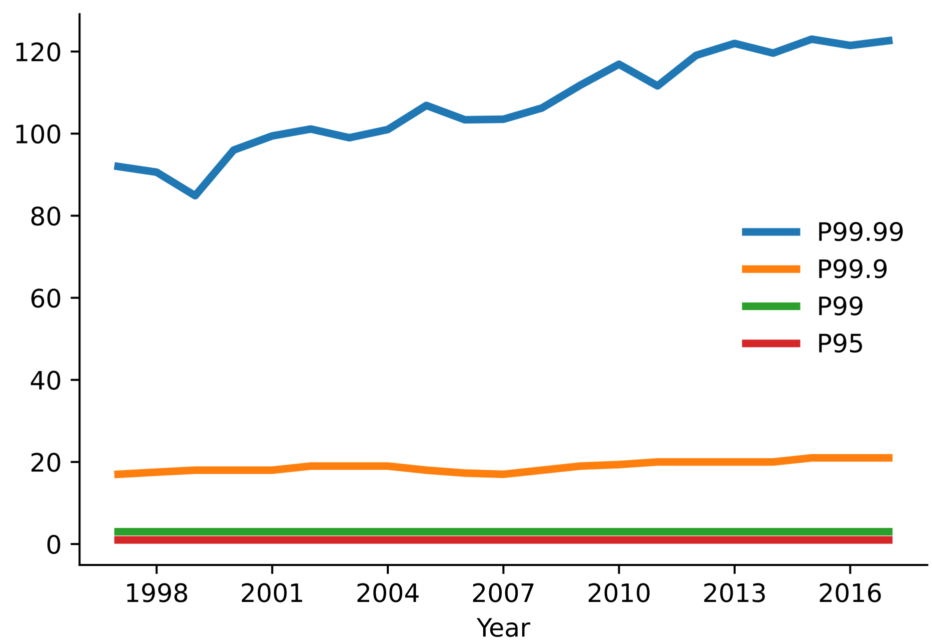
As shown in Figure 10, only a small fraction of firms operate more than one establishment.[18] Even the firm at the 95th percentile of the firm-size distribution (measured by the number of establishments) operates only a single establishment. Nonetheless, the number of establishments increases sharply in the upper tail: in 1997, firms at the 99th percentile operated three establishments, those at the 99.9th percentile operated 17, and those at the 99.99th percentile operated 92. Importantly, these extensive-margin measures of firm size increased substantially over time. By 2017, the 99.9th percentile had risen to 21 establishments and the 99.99th percentile to 123—representing increases of 24% and 34%, respectively.[19]

The addition of new establishments enables firms to reach new markets—whether in product space or geographic space—consistent with the mechanism emphasized in the model. Evidence from Cao et al. (2022) and Hsieh and Rossi-Hansberg (2023) similarly shows that firms increased their establishment count in U.S. Census data. Cao et al. (2022) further exploits this fact for identification in a model of creative destruction, likewise concluding that firm expansion costs have declined.

[18] Kondziella and Weiss (2025) provide a detailed characterization of multi-establishment firms in Swedish administrative data.

[19] The discreteness of the measure makes it unclear whether the observed stability for the lower percentiles is genuine or merely reflects that only extreme cost shifts would register as an additional establishment.

Figure 10: Distribution of establishments per firm



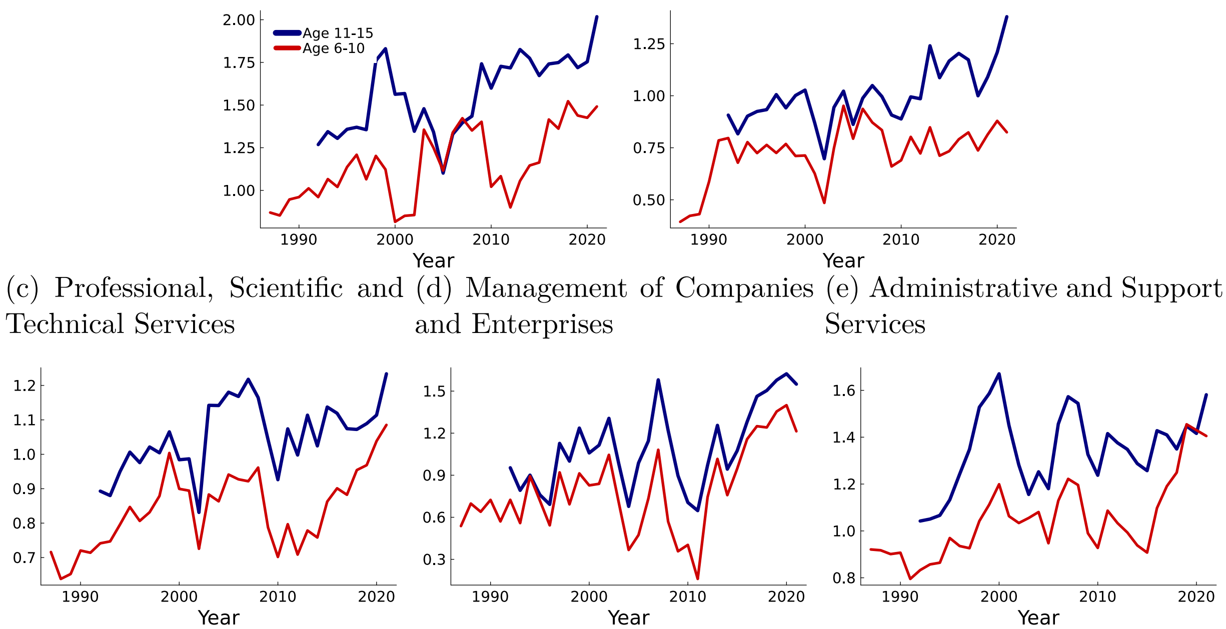
Notes: the figure shows the distribution of establishments per firm. The legend indicates percentiles of the distribution.

7.2 Firm size trends in the U.S.

Hopenhayn et al. (2022) and Karahan et al. (2024) document that firm size conditional on age has remained remarkably stable in U.S. Census data, for both entrants and mature firms. This stability implies that the gap in size between older firms and entrants—that is, the average firm-size trajectory conditional on survival—has changed little over time. Hence, at the aggregate level, U.S. Census data provide no clear evidence of a secular steepening in firm-size trajectories. At the sectoral level, however, some steepening is evident. Figure 11 plots, for selected service industries, the log average employment of firms aged six to ten and eleven to fifteen relative to entrants (age zero), using data from the U.S. Census Bureau's Business Dynamics Statistics (BDS). Each series is shown from its earliest available year through 2021.

Figure 11 shows that, over the past 30 years, the size gap between entrants and firms aged six to ten (eleven to fifteen) has widened in several service industries, including Information; Finance and Insurance; Professional, Scientific, and Technical Services; Management of Companies and Enterprises; and Administrative and Support Services. In the Information sector, for instance, firms aged eleven to fifteen expanded by roughly 0.75 log points relative to entrants (from a relative log size of 1.25 to 2). These patterns suggest that declining expansion costs are at work in at least these service sectors. Moreover, Figure 11 aligns with the evidence in Cao et al. (2022), who show using U.S. Census data that firms—particularly

Figure 11: Average employment by firm age relative to entrants in the U.S.



(a) Information (b) Finance and Insurance

(c) Professional, Scientific and Technical Services (d) Management of Companies and Enterprises (e) Administrative and Support Services

Notes: the figures show the average employment of firms aged six to ten and eleven to fifteen relative to the average employment of entrants (age zero) in logs by sector. The sectors correspond to the NAICS codes 51, 52, 54, 55 and 56. Data: U.S. Census Bureau - Center for Economic Studies - Business Dynamics Statistics (2021).

in services sectors—have expanded by adding new establishments. One possible reason why these trends are less visible at the aggregate level in the U.S. than in Sweden is that the share of firms operating in the sectors highlighted in Figure 11 is lower in the U.S. Census data than in the Swedish administrative data (23% vs. 28.4% in 2000).

7.3 Firm growth vs. selection

Is the steepening of the average size trajectory driven by faster firm growth, or by changes in the selection of surviving firms? This section disentangles these two margins using the structural model, showing that both margins shift differently across productivity types—giving rise to a steeper average trajectory.

Eq. (21) highlighted that the average firm-size trajectory is the weighted sum of productivity type-specific trajectories, where weights reflect the share of each type among surviving firms. Both components are determined by the type-specific expansion R&D rates. Higher expansion rates allow firms to enter more product markets, raising size growth conditional on survival and increasing survival itself. The transition-path analysis showed that expansion rates for both types increase in the short run; over the longer run, the rate for high-productivity firms continues to rise, while that for low-productivity firms gradually declines. Table 6 reports for both balanced growth paths the productivity-type specific expansion R&D rates, size-trajectories and survivor shares. For high-productivity firms, the

Table 6: Firm size growth and selection

	Initial BGP	New BGP
Expansion R&D rate (high-prod.) x^h	0.144	0.228
Expansion R&D rate (low-prod.) x^ℓ	0.091	0.08
Share (high-prod.) among surviving firms s^h	0.543	0.619
Growth cond. on survival (high-prod.)	0.577	0.842
Growth cond. on survival (low-prod)	0.391	0.307

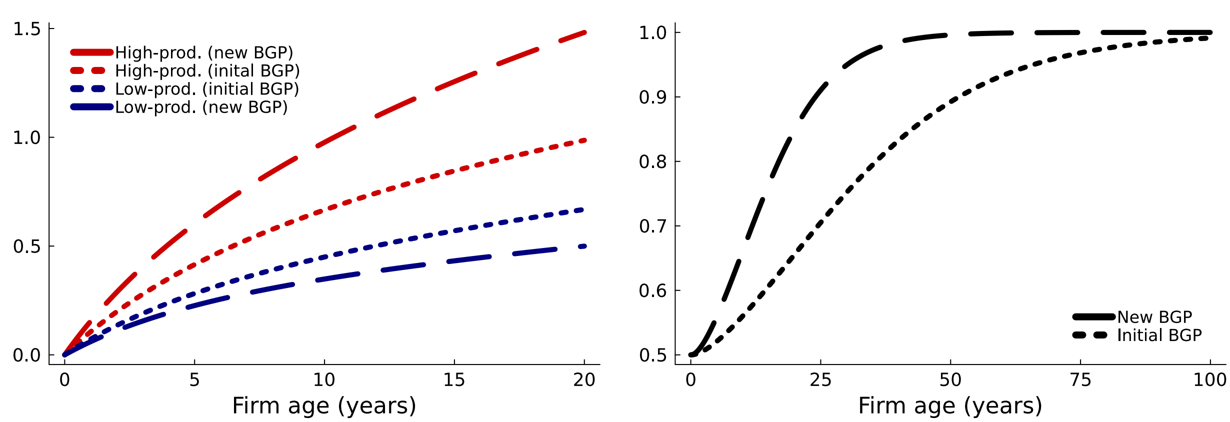
Notes: the table shows the expansion R&D rates of high- and low-productivity firms (first two rows), the share of high-productivity firms among firms alive at age eight (third row) and firm size growth (value added) over the first eight years for high- and low productivity firms in logs. Parameter values of the initial and new balanced growth path (BGP) are shown in Table 4.

expansion rate rises from 0.144 to 0.228, while for low-productivity firms it falls from 0.091 to 0.08. As a result, size growth (measured by value added) conditional on survival over the first eight years increases from 0.577 to 0.842 log points for high-productivity firms and decreases from 0.391 to 0.307 log points for low-productivity firms.[20] At the same time, higher expansion rates boost survival for high-productivity firms, while the opposite holds for low-productivity firms. Consequently, the share of high-productivity firms among eight-year-old survivors, $s^h(8)$, rises from 0.543 to 0.619. Therefore, eq. (21) implies that the steepening of the average size trajectory is driven by two forces: more productive firms not only grow faster conditional on survival, but also make up a larger share of the survivor pool.

Figure 12: Firm size growth and selection

(a) Size growth conditional on survival

(b) High-prod. type among surviving firms



Notes: the left figure shows firm size (value added) relative to entry (in logs) for high and low productivity firms as a function of firm age. The right figure shows the share of high productivity firms among surviving firms as a function of firm age.

Figure 12 illustrates this mechanism more generally. The left panel shows firm size growth (value added) conditional on survival for high- and low-productivity firms, while the right

[20] Consistent with these productivity-type–specific trends, Cao et al. (2022) show using U.S. Census data that large firms expanded in size while smaller firms contracted during the 1990s and 2000s.

panel plots the share of high-productivity firms among surviving firms, s^h, as a function of firm age. At age eight, the values align with those reported in Table 6. The increase (decrease) in the expansion R&D rates of high (low) productivity firms increases (decreases) firm growth conditional on survival. In the right panel, the share of high-productivity firms among survivors begins at 0.5 at age zero, reflecting the share of high-type entrants (p^h), and gradually converges to one. This convergence occurs because high-productivity firms, owing to their higher expansion R&D rates, are more likely to survive. In other words, although few firms survive to older ages, those that do are increasingly of the high-productivity type. As the gap in expansion R&D rates widens across balanced growth paths, this convergence accelerates, implying a higher share of high-productivity survivors at every firm age.

7.4 Alternative parameterizations of p^h

In the main analysis, the parameter p^h capturing the share of high-productivity firms among entrants was set exogenously to 0.5. This value was motivated by the fact that the interquartile range captures size dispersion around the median firm. I relax this assumption in this section. In particular, I estimate the initial and new balanced growth path for alternative parameterizations of p^h and, as before, quantify the contributions of all parameters to the changes in the targeted moments each time.

Table 7 shows the estimated parameters in the initial and new balanced growth path for two alternative parameterizations of p^h, in particular $p^h = 0.4$ and $p^h = 0.6$. The results are similar to before. In particular, both alternative estimations ask for a significant increase in the expansion R&D efficiency. One notable observation is that the estimation asks for a larger productivity gap in the initial balanced growth path when $p^h = 0.6$ than when $p^h = 0.4$. In other words, the model implies a greater productivity differential between the top 60% and bottom 40% of entrants than between the top 40% and bottom 60%. Likewise, the increase in the productivity gap across balanced growth paths is more pronounced when p^h is set to 0.6. For p^h set to 0.4, the estimation even asks for a slight decrease in the productivity gap. Therefore, the estimation suggests that productivity dispersion increased primarily between the top 60% and bottom 40% of entrants, rather than between the top 40% and bottom 60%. This finding is notable, as the existing literature has instead emphasized the role of top, so-called 'superstar', firms in accounting for recent macroeconomic trends.

Table 8 quantifies the contribution of each parameter to the changes in the targeted moments for both alternative parameterizations of p^h. As before, the decomposition follows eq. (24) and the notation Table 5. The results are quantitatively in line with the previous parameterizations: the increase in the expansion R&D efficiency explains the changes in the average firm-size trajectory and cross-sectional size dispersion (columns VA, wL and σ_{VA}) jointly.

Table 7: Estimation for alternative values of p^h

	Initial BGP		New BGP	
	Data	Model	Data	Model
$\boldsymbol{p^h = 0.4}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.637
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.623
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
Entry rate in %	14.8	14.8	11	11
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.894		2.065
ψ_I *Internal R&D efficiency*		0.936		1.357
φ^h/φ^ℓ *Productivity gap*		1.017		1.014
ψ_z *Entry R&D efficiency*		3.786		6.329
λ *Step size of quality improvements*		1.051		1.035
$\boldsymbol{p^h = 0.6}$				
Moments				
Value added at age 8 relative to entry in logs	0.492	0.492	0.638	0.649
Wage bill at age 8 relative to entry in logs	0.468	0.468	0.623	0.632
Interquartile range of log value added	log(2)	log(2)	log(5)	log(5)
Entry rate in %	14.8	14.8	11	11
TFP growth g in %	1.5	1.5	1.1	1.1
Parameters				
ψ_x *Expansion R&D efficiency*		0.888		1.814
ψ_I *Internal R&D efficiency*		0.905		0.913
φ^h/φ^ℓ *Productivity gap*		1.022		1.038
ψ_z *Entry R&D efficiency*		3.792		6.522
λ *Step size of quality improvements*		1.051		1.038

7.5 Discussion on markups

Evidence on markups for the Swedish economy is scarce. Sandström (2020) provides an estimate of the aggregate markup using, as I do, administrative data on the balance sheets of Swedish firms. The author reports an estimate of about 15% for the sales-weighted aggregate markup and 5% for the unweighted one. These estimates are lower than common markup estimates for the U.S. (De Loecker et al., 2020); however, a direct comparison is difficult due to differences in, e.g., the data coverage or the institutional setting. Importantly, the markup estimates in Sandström (2020) are stable, displaying no systematic trend over time. This stability motivates the exclusion of the aggregate markup from the main analysis.[21]

[21] In the model, the (cost-weighted) markup in the estimated initial balanced growth path stands at 7.1%, well within the estimates provided in Sandström (2020) and even declines slightly in the new balanced growth

Table 8: Decomposition for alternative values of p^h

	VA	wL	σ_{VA}	$\frac{z}{M}$	g
$\boldsymbol{p^h = 0.4}$					
Expansion R&D efficiency $\psi_x \uparrow$	380.3	364.7	108.4	184.4	-39.6
Internal R&D efficiency $\psi_I \uparrow$	3.7	-2.2	-4.7	-31.2	-54.8
Productivity gap $\varphi^h/\varphi^\ell \downarrow$	-3.2	-2.7	-2.8	-7.5	-1.4
Entry R&D efficiency $\psi_z \uparrow$	-126.5	-115.4	-40.3	-346.0	-94.4
Step size of quality improvements $\lambda \downarrow$	8.5	12.4	33.1	194.8	201.7
$\boldsymbol{p^h = 0.6}$					
Expansion R&D efficiency $\psi_x \uparrow$	281.0	275.0	95.7	128.0	-33.9
Internal R&D efficiency $\psi_I \uparrow$	0.1	-0.1	-0.1	-0.6	-1.1
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	24.3	21.3	25.5	51.5	9.2
Entry R&D efficiency $\psi_z \uparrow$	-120.5	-112.0	-49.3	-347.8	-103.6
Step size of quality improvements $\lambda \downarrow$	6.5	9.7	30.3	147.6	163.5

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (24). Moments and parameter values of the initial and new BGP are shown in Table 7. VA denotes the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σ_{VA} the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

8 Conclusion

Cross-sectional firm-size dispersion can increase because average firm size rises more steeply with age, because the distribution of firms shifts toward older cohorts, or because size heterogeneity conditional on age expands. Using Swedish administrative data, this paper documents that the size–age relationship has steepened over time in parallel with rising cross-sectional size dispersion. Embedding this finding in a model of creative destruction with rich firm dynamics reveals that declining costs of expanding into new product markets are the primary mechanism—raising size dispersion by steepening the average firm-size trajectory. The model further shows that alternative forces—higher entry costs, which primarily shift the firm age distribution, and rising productivity dispersion, which mainly increase size heterogeneity conditional on age—have played more limited roles in the overall rise in size dispersion.

What real-world forces might the decline in expansion costs reflect? New technologies have enabled firms to manage and operate multiple production units more efficiently, effectively reducing the organizational costs of scaling up (Hsieh and Rossi-Hansberg, 2023). These advances plausibly stem from the ICT revolution around the turn of the millennium, which aligns closely with the timing of the changes in firm-size dynamics documented here. Future

path. Despite high-productivity firms with relatively high markups capturing additional market shares in the new balanced growth path, the fall in the step size of quality improvements puts downward pressure on within-firm markups.

research could provide direct empirical evidence on this mechanism. Understanding the source of falling expansion costs is important, given its central role in explaining the rise of firm-size dispersion.

The results of the paper have broader implications. First, many existing narratives emphasize firm aging as a key driver of recent macroeconomic trends. This paper instead highlights that productive incumbents' active expansion—rather than firms' passive aging—is central to understanding rising market concentration. Second, lower expansion costs pose a tradeoff: they promote growth at the firm level but also contribute to increasing concentration in the macroeconomy. Antitrust policy should limit the ex-post losses arising from the market power of large incumbents while preserving growth incentives for young firms.

Lastly, Section 7.2 identified several U.S. services industries in which the average firm-size trajectory has steepened. Notably, the share of firms operating in these sectors has risen from 15.4% in 1978 to 24.4% in 2023 in the Census data. This suggests that, if expansion costs continue to fall and these services sectors keep gaining importance, these trends will increasingly shape firm-size dynamics at the aggregate level in the U.S. going forward.

References

Acemoglu, D., Akcigit, U., Alp, H., Bloom, N. and Kerr, W. (2018), 'Innovation, reallocation, and growth', *American Economic Review* **108**(11), 3450–91.

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J. and Li, H. (2023), 'A theory of falling growth and rising rents', *Review of Economic Studies* **90**(6), 2675–2702.

Aghion, P. and Howitt, P. (1992), 'A Model of Growth through Creative Destruction', *Econometrica* **60**(2), 323–351.

Akcigit, U. and Ates, S. T. (2021), 'Ten facts on declining business dynamism and lessons from endogenous growth theory', *American Economic Journal: Macroeconomics* **13**(1), 257–98.

Akcigit, U. and Kerr, W. R. (2018), 'Growth through heterogeneous innovations', *Journal of Political Economy* **126**(4), 1374–1443.

Autor, D., Dorn, D., Katz, L. F., Patterson, C. and Van Reenen, J. (2020), 'The fall of the labor share and the rise of superstar firms', *The Quarterly Journal of Economics* **135**(2), 645–709.

Bloom, N., Jones, C. I., Van Reenen, J. and Webb, M. (2020), 'Are ideas getting harder to find?', *American Economic Review* **110**(4), 1104–1144.

Blundell, R., Griffith, R. and Windmeijer, F. (2002), 'Individual effects and dynamics in count data models', *Journal of econometrics* **108**(1), 113–131.

Cao, D., Hyatt, H. R., Mukoyama, T. and Sager, E. (2022), 'Firm growth through new establishments'.

Chen, Y.-F., Hsu, W.-T. and Peng, S.-K. (2023), 'Innovation, firm size distribution, and gains from trade', *Theoretical Economics* **18**(1), 341–380.

Chiavari, A. and Goraya, S. (2020), The rise of intangible capital and the macroeconomic implications, Technical report, Working paper.

Davis, S. J. (2017), 'Regulatory complexity and policy uncertainty: headwinds of our own making', *Becker Friedman Institute for Research in economics working paper* (2723980).

De Loecker, J., Eeckhout, J. and Unger, G. (2020), 'The rise of market power and the macroeconomic implications', *The Quarterly Journal of Economics* **135**(2), 561–644.

De Ridder, M. (2024), 'Market power and innovation in the intangible economy', *American Economic Review* **114**(1), 199–251.

Decker, R. A., Haltiwanger, J., Jarmin, R. S. and Miranda, J. (2016), 'Declining business dynamism: What we know and the way forward', *American Economic Review* **106**(5), 203–07.

Decker, R. A., Haltiwanger, J., Jarmin, R. S. and Miranda, J. (2020), 'Changing business dynamism and productivity: Shocks versus responsiveness', *American Economic Review* **110**(12), 3952–90.

Engbom, N. (2023), 'Misallocative growth', *Working Paper* .

Gourio, F., Messer, T. and Siemer, M. (2014), A missing generation of firms? aggregate effects of the decline in new business formation, *in* 'CES Ifo Conference'.

Grullon, G., Larkin, Y. and Michaely, R. (2019), 'Are us industries becoming more concentrated?', *Review of Finance* **23**(4), 697–743.

Gutiérrez, G. and Philippon, T. (2018), *How EU markets became more competitive than US markets: A study of institutional drift*, number w24700, National Bureau of Economic Research New York,.

Hall, B. H. and Ziedonis, R. H. (2001), 'The patent paradox revisited: an empirical study of patenting in the us semiconductor industry, 1979-1995', *rand Journal of Economics* pp. 101–128.

Hopenhayn, H., Neira, J. and Singhania, R. (2022), 'From population growth to firm demographics: Implications for concentration, entrepreneurship and the labor share', *Econometrica* **90**(4), 1879–1914.

Hsieh, C.-T. and Rossi-Hansberg, E. (2023), 'The industrial revolution in services', *Journal of Political Economy Macroeconomics* **1**(1), 3–42.

Karahan, F., Pugsley, B. and Şahin, A. (2024), 'Demographic origins of the start-up deficit', *American Economic Review* **114**(7), 1986–2023.

Klette, T. J. and Kortum, S. (2004), 'Innovating firms and aggregate innovation', *Journal of political economy* **112**(5), 986–1018.

Kondziella, M. and Weiss, J. (2025), ‘More than the sum of its parts? markups and the role of establishments’.

Lentz, R. and Mortensen, D. T. (2008), ‘An empirical model of growth through product innovation’, *Econometrica* **76**(6), 1317–1373.

Olmstead-Rumsey, J. (2019), ‘Market concentration and the productivity slowdown’.

Peters, M. (2020), ‘Heterogeneous markups, growth, and endogenous misallocation’, *Econometrica* **88**(5), 2037–2073.

Peters, M. and Walsh, C. (2021), Population growth and firm dynamics, Technical report, National Bureau of Economic Research.

Sandström, M. (2020), *Intangible Capital, Markups and Profits*.

Weiss, J. (2019), Intangible investment and market concentration, Technical report, Working paper.

Online Appendix for

"Recent Changes in Firm Dynamics and the Nature of Macroeconomic Trends"

[FOR ONLINE PUBLICATION ONLY]

A Data

The main data set, *Företagens Ekonomi* (FEK), covers information from balance sheets and profit and loss statements for the universe of Swedish firms. From this data, I obtain the main variables of interest, namely value added (*Förädlingsvarde*, variable name: *Foradlingsvarde*) and the wage bill (*Lönekostnader*, variable name: *KonstnaderForLonerOchAndraErsattn*). In the FEK code-book by Statistics Sweden (SCB), these variables are defined as follows.[22] SCB defines value added as a firm's production value minus costs for purchased goods and services used as inputs in production (salaries and social security contributions not included). SCB's measure of value added captures the firm's contribution to the gross domestic product. The measure of the firm's wage bill includes salaries and other remuneration, including severance pay. As described in the main text, I focus on firms in the private sector. These firms have a legal type (variable name: *JurForm*) less than 50 or equal to 96.

To measure firm-size dispersion within industries, I use the industry classification (SNI codes) provided by SCB. The industry classification changed twice between 1997 and 2017, once in 2002 and once in 2007. The changes in industry classification mainly affect the more detailed codes at the five-digit level, whereas I use the ones at the two-digit level for the main analysis. Nevertheless, to ensure a consistent industry classification over time I proceed as follows.

[22] `https://www.scb.se/contentassets/9dd20ce462644cc19f6f04eb2edbbe28/nv0109_kd_slut2022_v1_20240510.pdf`, accessed 26.03.2025.

During the year of the change, I observe both the old and the new industry classifications. For the firms present in the data in the year of the classification change, extending the new industry classification further back in time before the change is straightforward. This way, the industry codes of almost all firms are updated. A firm might be in the data before and after the classification change but not for the year of the change. For these firms, the above method does not work. If the firm appears in the data one year after the classification change, I use the observed classification after the change to update the classification before the change. For firms that are absent for several years around the year of change, I use industry mappings provided by Statistics Sweden. These mappings do not always provide a 1:1 mapping between industries before and after the classification change, so I use the most common transitions for the m:m mappings.

One concern is that changes in the firm structure, e.g., when firms merge with other firms, change the firm ID. To address this concern, I impute changes in firm IDs using worker flows between firms. The auxiliary data set *Registerbaserad Arbetsmarknadsstatistik* (RAMS) contains the universe of employer-employee matches. I impute changes in the firm ID of firms with at least five employees as follows: if more than 50% of the workforce of firm A in year t makes up for more than 50% of the workforce of firm B in year $t+1$, I substitute firm B's firm ID by firm A's firm ID following $t+1$. The empirical results remain unchanged when excluding firms for whom the imputed firm ID differs from the observed firm ID.

Other data sources

- U.S. Census Bureau - Center for Economic Studies - Business Dynamics Statistics (2021). Accessed July 13, 2024.
- Federal Reserve Economic Data (FRED): Total Factor Productivity at Constant National Prices for Sweden (series RTFPNASEA632NRUG). Accessed Tuesday, January 30, 2024.

B Trends in the Swedish economy

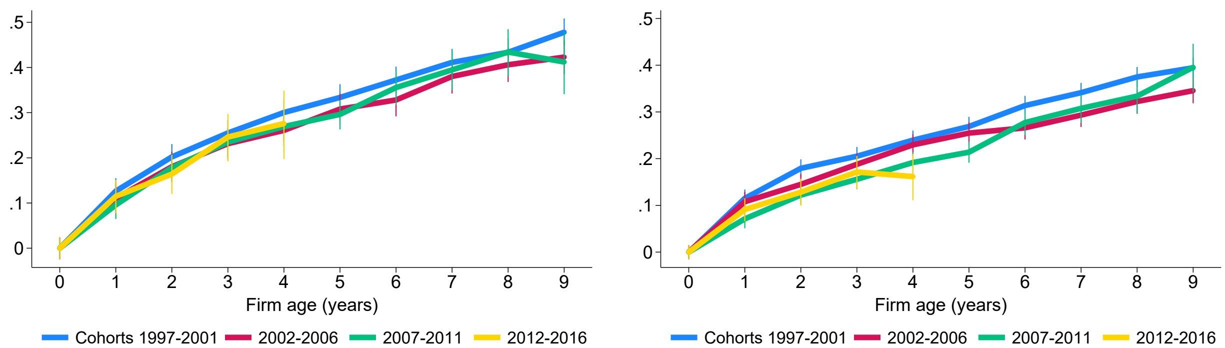
B.1 Trends in the dynamics of firm size

This section documents additional trends in firm size in the Swedish economy. Figure A-1a shows the average physical capital stock relative to entry by cohort (in logs). The capital stock is defined as fixed assets minus depreciation. The different cohort trajectories overlap, indicating no systematic difference in firms' capital accumulation over time. This suggests that the increase in firm size relative to entry documented in the main text is not driven by capital deepening.

Figure A-1: Capital stock relative to entry

(a) Physical capital

(b) Physical capital plus intangibles

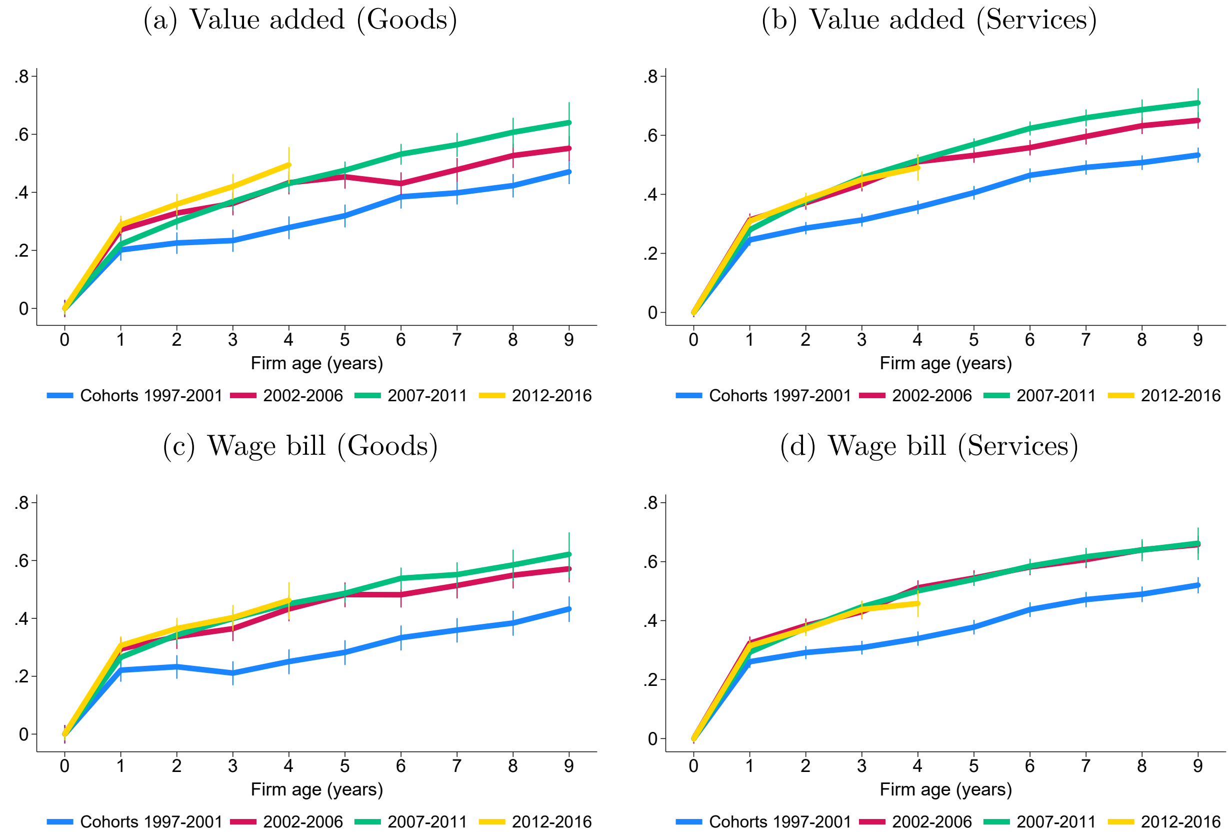


Notes: the figure shows the average capital stock of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The left panel captures physical capital measured as fixed assets minus depreciation and the right panel adds intangible assets to the capital measure. The cohort 2004 is excluded as explained in the main text. 95% confidence bands are included.

This picture does not change even when including intangible assets in the measure of the capital stock. Figure A-1b repeats the previous graph for the measure of capital that includes firms' (book value of) intangible capital. The trajectories become slightly more dispersed compared to Figure A-1a; however, there is no systematic trend across cohorts visible.

Next, I show the size trajectories of value added and the wage bill by sector. The left column in Figure A-2 is restricted to firms in the goods sector, and the right column to firms in the services sector. The trends are visible in both sectors, particularly in the services sector. That the trends are particularly clear for services firms whose products generally are locally provided and consumed suggests that the trends in firm size are not driven by foreign goods demand.

Figure A-2: Firm-size relative to entry (by sector)



Notes: the figure shows the average value added and wage bill of surviving firms relative to entry (in logs) for different cohorts as indicated in the legend. The left column is restricted to firms in the goods sector and the right column to firms in the services sector. 95% confidence bands are included.

Lastly, I assess the robustness of the steepening of the firm-size trajectories using a regression framework. Specifically, I regress firm size (measured by value added or the wage bill) on firm age, controlling for interacted cohort and five-digit industry fixed effects. Formally,

$$Size_{f,t} = \beta_0 + \beta_1 Age_{f,t} + \delta_{c,n} + \epsilon_{f,t}, \tag{A-1}$$

where c and n index cohorts and industries, respectively. I estimate the regression in levels rather than logarithms, since value added can be negative and the wage bill is zero for self-employed entrepreneurs without employees, who are included in the data. Excluding such firm-year observations would introduce selection among firm-size trajectories as entrepreneurs commonly start businesses without any labor costs. To study how the size-age relationship evolves over time, I estimate the regression separately by cohort groups and focus on firms' first five years of life. I restrict attention to cohorts founded before 2013 to ensure each is observed for a full five-year window.

Table A-1 presents the regression results using value added as the measure of firm size. The coefficient on firm age increases noticeably over time. For firms established between 1997 and 1999, each additional year of age is associated, on average, with an increase in value added of 140,468 SEK (in 2017 prices; roughly 14,000 USD). This effect rises to 189,504 SEK for firms founded between 2000 and 2002, 195,526 SEK for those founded between 2003 and 2005, and 193,044 SEK for firms established thereafter. These results confirm not only the steepening of the size–age relationship, but also that this trend was particularly pronounced for cohorts founded around the turn of the millennium. Table A-2 shows very similar patterns when using the wage bill as the size measure.

Table A-1: Size-age regressions: value added

	Size	Size	Size	Size
Age	140,468	189,504	195,526	193,044
	(4,713)	(4,916)	(5,339)	(2,670)
Cohort × industry fixed effects	✓	✓	✓	✓
Cohorts 1997-1999	✓			
Cohorts 2000-2002		✓		
Cohorts 2003-2005			✓	
Cohorts 2006-2012				✓
R^2	0.119	0.148	0.129	0.108
Number of observations	321,027	321,588	212,451	873,447

Notes: the table shows the results of the regression in eq. (A-1) using value added as the firm-size measure. The age coefficient is in units of 2017 SEK (1 SEK ≈ 0.1 USD). All regressions control for interacted cohort and five-digit industry fixed effects. Standard errors are clustered at the firm level. The 2004 cohort is excluded as explained in the main text.

Table A-2: Size-age regressions: wage bill

	Size	Size	Size	Size
Age	79,321	93,508	108,851	109,902
	(2,638)	(2,709)	(3,013)	(1,565)
Cohort × industry fixed effects	✓	✓	✓	✓
Cohorts 1997-1999	✓			
Cohorts 2000-2002		✓		
Cohorts 2003-2005			✓	
Cohorts 2006-2012				✓
R^2	0.122	0.150	0.134	0.115
Number of observations	321,027	321,588	212,451	873,447

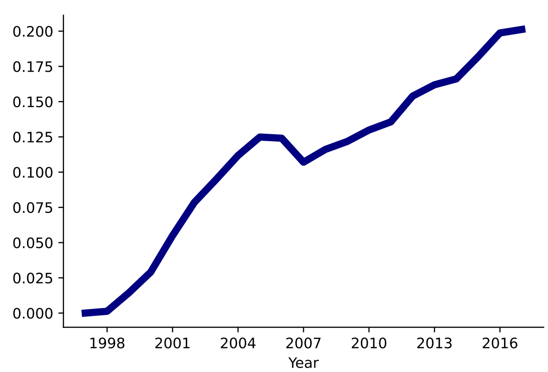
Notes: the table shows the results of the regression in eq. (A-1) using firms' wage bills as the size measure. The age coefficient is in units of 2017 SEK (1 SEK ≈ 0.1 USD). All regressions control for interacted cohort and five-digit industry fixed effects. Standard errors are clustered at the firm level. The 2004 cohort is excluded as explained in the main text.

B.2 Trends in the macroeconomy

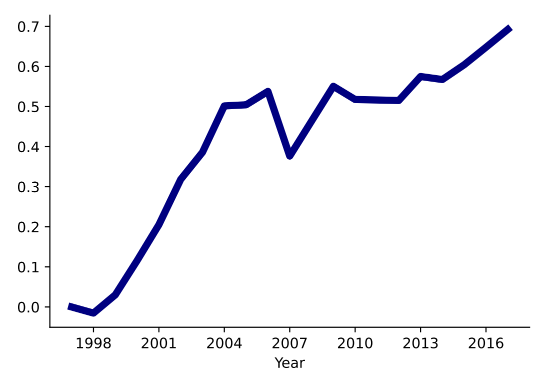
This section presents the evolution of the two main measures of cross-sectional firm-size dispersion—namely, the standard deviation and interquartile range of log value added—using finer, five-digit industry classifications. The benefit of this approach is that it compares firms operating in more narrowly defined product markets. The drawback is that, at this level of granularity, some industries have a negative 25th percentile of value added, making the logarithm undefined. For the interquartile range in Figure A-3b, I exclude such product groups from the analysis. For the standard deviation in Figure A-3a, all firm-year observations with negative value added are excluded as in the main analysis. Despite this limitation, the evolution of firm-size dispersion closely mirrors the patterns documented in the main text, as Figure A-3 illustrates.

Figure A-3: Cross-sectional firm size dispersion

(a) Standard deviation



(b) Interquartile range



Notes: the left panel shows the standard deviation and the right panel the interquartile range of log value added. Both measures are computed within five-digit industries and aggregated using industry labor costs. The value in 1997 is normalized to zero.

C Model

C.1 Solving the dynamic firm problem

The HJB for a high productivity-type firm h reads[23]

$$
\begin{aligned}
&r_t V_t^h(n,[\mu_i],S_t) - \dot{V}_t^h(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^h\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^h\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^h\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) + (1-S_t) V_t^h\Big(n+1,\Big[[\mu_i],\lambda\times\varphi^h/\varphi^l\Big],S_t\Big) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^{\zeta} + \frac{1}{\psi_x}(x_k)^{\zeta} \right] \Bigg\}
\end{aligned}
$$

The HBJ for a low productivity-type firm l reads

$$
\begin{aligned}
&r_t V_t^l(n,[\mu_i],S_t) - \dot{V}_t^l(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^l\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \Bigg\{ \sum_{k=1}^{n} I_k \Big[V_t^l\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^l\Big(n+1,\Big[[\mu_i],\lambda\times\varphi^l/\varphi^h\Big],S_t\Big) + (1-S_t) V_t^l\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^{\zeta} + \frac{1}{\psi_x}(x_k)^{\zeta} \right] \Bigg\}.
\end{aligned}
$$

I solve for the value function of a high-type firm, however the steps for the low-type firm are equivalent. For clarity, I suppress the dependence of the value function on S_t in the following. Guess that the value function of the firm consists of a component that is common to all lines and a line-specific component

$$V_t^h(n,[\mu_i]) = V_{t,P}^h(n) + \sum_{k=1}^{n} V_{t,M}^h(\mu_k).$$

[23] The derivation follows Peters (2020) when applicable.

Substituting the guess into the HJB, $V^h_{t,P}(n)$ and $V^h_{t,M}(\mu_k)$ solve the following differential equations

$$\begin{aligned} r_t V^h_{t,M}(\mu_i) - \dot{V}^h_{t,M}(\mu_i) =& \pi(\mu_i) - \tau_t V^h_{t,M}(\mu_i) \\ &+ \max_{I_i} \left\{ I_i \Big[V^h_{t,M}(\mu_i \times \lambda) - V^h_{t,M}(\mu_i) \Big] - w_t \mu_i^{-1} \frac{1}{\psi_I} (I_i)^\zeta \right\} \end{aligned} \tag{A-2}$$

and

$$\begin{aligned} & r_t V^h_{t,P}(n) - \dot{V}^h_{t,P}(n) = \sum_{k=1}^{n} \tau_t \Big[V^h_{t,P}(n-1) - V^h_{t,P}(n) \Big] \\ &+ \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \Big[V^h_{t,P}(n+1) - V^h_{t,P}(n) + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) \Big] - w_t \frac{1}{\psi_x} (x_k)^\zeta \right\}. \end{aligned} \tag{A-3}$$

Assume that in steady-state $V^h_{t,P}$ and $V^h_{t,M}$ grow at the constant rate g. Using this guess in eq. (A-2) and following Peters (2020), we obtain for $V^h_{t,M}(\mu_i)$

$$V^h_{t,M}(\mu_i) = \frac{\pi(\mu_i) + \frac{\zeta - 1}{\psi_I} (I_i)^\zeta w_t \mu_i^{-1}}{\rho + \tau},$$

where I_i solves

$$I_i = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta - 1}{\psi_I} (I_i)^\zeta \right) \left(1 - \frac{1}{\lambda} \right) \frac{\psi_I}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}. \tag{A-4}$$

Eq. (A-4) shows that internal innovation rates I_i are time invariant, and independent of the product line and the productivity type of the firm, $I \equiv I^h = I^l$.

With this at hand, we can turn back to the differential equation for $V^h_{t,P}(n)$ in eq. (A-3). In addition to the guess that $V^h_{t,P}(n)$ grows at rate g, conjecture that $V^h_{t,P}(n) = n \times v^h_t$. Combined with the Euler we get

$$(\rho + \tau) n v^h_t = \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \Big[v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) \Big] - w_t \frac{1}{\psi_x} (x_k)^\zeta \right\}. \tag{A-5}$$

The optimality condition for x_k is given by

$$v^h_t + S_t V^h_{t,M}(\lambda) + (1-S_t) V^h_{t,M}(\lambda \times \varphi^h/\varphi^l) = w_t \frac{\zeta}{\psi_x} (x_k)^{\zeta - 1}. \tag{A-6}$$

Several observations are noteworthy. First, eq. (A-6) shows that optimal expansion rates are independent of quality and productivity gaps in line k. We can hence drop the item indexation: $x_k = x^d$, where $d \in \{h, \ell\}$. Second, $v_t, V^h_{t,M}, w_t$ all grow at the same rate g, which implies that expansion rates are constant over time. We can hence write eq. (A-5) as

$$v^h_t = \frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^h)^\zeta w_t.$$

Gathering all terms, the value function is given by

$$\begin{aligned}
V^h_t(n,[\mu_i]) =& V^h_{t,P}(n) + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\
=& n v^h_t + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\
=& n\frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^h)^\zeta w_t + \sum_{k=1}^{n}\frac{\pi(\mu_k)+\frac{\zeta-1}{\psi_I}I^\zeta w_t \mu_k^{-1}}{\rho+\tau},
\end{aligned} \tag{A-7}$$

which is the expression for the value function stated in the main text, Proposition 1.

Using the expression for v^h_t, write the optimality condition in eq. (A-6) as

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^l}{\varphi^h}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1}.$$

Following the same steps for low-productivity firms, we obtain the optimality condition

$$\frac{\zeta-1}{\psi_x}(x^l)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\frac{\varphi^h}{\varphi^l}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\lambda^{-1}\right)$$
$$= (\rho+\tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta-1}.$$

Lastly, I prove that more productive firms choose higher expansion R&D rates, i.e., $x^h > x^l$. Intuitively, the proof shows that an increase in productivity raises the stream of profits in a product line. The continuation value and the marginal cost of expansion R&D have to rise for the optimality condition of the expansion R&D rate to hold, implying that the expansion R&D rate increases. First note that product markups are increasing in firm productivity, as shown in eq. (4). Next, I totally differentiate the optimality condition for the expansion

R&D rate and show that the expansion R&D rate is increasing in the markup.

Write the optimality condition for expansion R&D as[24]

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^h}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^h}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^l}\right)$$
$$=(\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1},$$

where μ^h and μ^l denote the initial markup charged when facing a high- and low-productivity firm. Totally differentiate with respect to markups and the expansion R&D rate

$$S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left((\rho+\tau)(x^h)^{\zeta-2}-(x^h)^{\zeta-1}\right)dx^h,$$

where $d\mu^h = d\mu^l \equiv d\mu$. Since $x^h > 0$, the above can be rearranged to

$$\frac{1}{(x^h)^{\zeta-2}}\left[S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)\right]d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left(\rho+\tau-x^h\right)dx^h.$$

The left hand side captures the effect of changes in markups on profits. The right hand side captures the effect of changes in the expansion R&D rate on the continuation value and marginal costs of expansion R&D. From eq. (A-4) we know that $\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta > 0$, otherwise the optimal internal R&D rate is negative. For a stationary firm size distribution, we must further have $\tau > x^h$. From this, it follows that $dx^h/d\mu > 0$, which concludes the proof.

C.2 Joint distribution of quality and productivity gaps

The distribution of quality and productivity gaps between incumbent and laggard firms characterizes economic aggregates in the model. On the one hand, quality and productivity gaps define product markups that determine labor demand. On the other hand, the joint distribution characterizes the share of product lines operated by each productivity type, which is a state variable in the firm's optimization problem. This section derives the joint distribution of quality and productivity gaps as a function of firm policies, which allows the equilibrium distribution to be solved jointly with the policies.

[24]This uses the optimality condition of the high-type firm but the one of the low type works equivalently.

The distribution of quality and productivity gaps across product lines is characterized by a set of infinitely many differential equations. For simplicity, I characterize the differential equations for firm-type specific expansion R&D rates, x_t^h and x_t^ℓ, and homogeneous internal R&D rates, I_t, as proven shortly in Proposition 1 for a balanced growth path.[25] The measure ν_t of product lines in which an incumbent of productivity type φ_f is Δ quality steps ahead of a $\varphi_{f'}$ type laggard follows (for $\Delta \geq 2$)

$$\dot{\nu}_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right) = I_t\nu_t\left(\Delta - 1, \frac{\varphi_f}{\varphi_{f'}}\right) - \nu_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right)(I_t + \tau_t). \tag{A-8}$$

For product lines where the incumbent is one step ahead ($\Delta = 1$), this measure follows

$$\begin{aligned}
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^h}\right) &= (1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t - \nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^l}\right) &= (1 - S_t)x_t^l(1 - S_t) + z_t(1 - p^h)(1 - S_t) - \nu_t\left(1, \frac{\varphi^l}{\varphi^l}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^h}\right) &= S_t x_t^h S_t + z_t p^h S_t - \nu_t\left(1, \frac{\varphi^h}{\varphi^h}\right)(I_t + \tau_t) \\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^l}\right) &= S_t x_t^h(1 - S_t) + z_t p^h(1 - S_t) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right)(I_t + \tau_t).
\end{aligned} \tag{A-9}$$

Changes in ν_t are due to inflows and outflows. Outflows from state Δ arise from successful internal R&D (quality gap increases from Δ to $\Delta + 1$) and creative destruction (Δ is reset to unity), as captured by the terms following the minus signs. Inflows vary with the quality gap. For $\Delta \geq 2$, inflows into state Δ are due to successful internal R&D in product lines with quality gaps of $\Delta - 1$. For $\Delta = 1$, inflows result from creative destruction. For example, the measure of products with a low-type incumbent and high-type laggard $\nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)$ increases due to low-type incumbents and entrants replacing high-type incumbents, captured by $(1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t$ in eq. (A-9).

The following characterizes the two-dimensional distribution of quality and productivity gaps along the BGP. I solve for the steady state distribution over quality and productivity gaps by setting the differential equations characterizing the law-of-motion in eq. (A-8) and (A-9) equal to zero. From this, one obtains the stationary mass of product lines with quality gap

[25] The distribution along the transition path is characterized in the Appendix, Section D.

λ^{Δ} and productivity gap φ^i/φ^j

$$\begin{aligned}
\nu\left(\Delta, \frac{\varphi^l}{\varphi^h}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{(1-S)x^l S + z(1-p^h)S}{I} \\
\nu\left(\Delta, \frac{\varphi^l}{\varphi^l}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{(1-S)x^l(1-S) + z(1-p^h)(1-S)}{I} \\
\nu\left(\Delta, \frac{\varphi^h}{\varphi^h}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{Sx^h S + zp^h S}{I} \\
\nu\left(\Delta, \frac{\varphi^h}{\varphi^l}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta} \frac{Sx^h(1-S) + zp^h(1-S)}{I}.
\end{aligned}$$

Summing over all Δ for a given productivity gap gives $S_{\varphi^l,\varphi^h}, S_{\varphi^l,\varphi^l}, S_{\varphi^h,\varphi^h}, S_{\varphi^h,\varphi^l}$ as stated in Proposition 1 in the main text. It follows that

$$\begin{aligned}
\Pr\left(\Delta \leq d, \frac{\varphi^l}{\varphi^h}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^{d}\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^l}{\varphi^l}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^l}{\varphi^l}\right) = S_{\varphi^l,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^{d}\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^h}{\varphi^h}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^h}\right) = S_{\varphi^h,\varphi^h}\left(1 - \left(\frac{I}{I+\tau}\right)^{d}\right) \\
\Pr\left(\Delta \leq d, \frac{\varphi^h}{\varphi^l}\right) &= \sum_{i=1}^{d} \nu\left(i, \frac{\varphi^h}{\varphi^l}\right) = S_{\varphi^h,\varphi^l}\left(1 - \left(\frac{I}{I+\tau}\right)^{d}\right).
\end{aligned}$$

Focusing on product lines where a low-productivity incumbent faces a high-productivity second-best firm:

$$P\left(\Delta \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-d[\ln(I+\tau) - \ln I]}\right)$$

or

$$P\left(\ln\left(\lambda^{\Delta}\right) \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - e^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}d}\right).$$

Conditional on the productivity gap, $\ln\left(\lambda^{\Delta}\right)$ is exponentially distributed with parameter

$\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Further

$$P\left(\lambda^{\Delta} \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - d^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter $\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Denote $\theta = \frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. We then have

$$P\left(\lambda^{\Delta} \leq m, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - m^{-\theta}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter θ. As in Peters (2020), θ is affected by the rate of internal R&D I relative to creative destruction τ. The higher the rate of internal R&D, the more mass is in the tail of the quality gap distribution. The difference to Peters (2020) is that, in this model, quality gaps *conditional* on the productivity gap are Pareto distributed.

The following uses the obtained results to solve for the aggregate labor income share, the misallocation measure and the aggregate markup. After repeating the same steps for lines with different productivity gaps, we obtain the aggregate labor income share as follows[26]

$$\begin{aligned}\Lambda =& \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{1}{\varphi_k/\varphi_n}\frac{1}{m}S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\ =&\frac{\theta}{\theta+1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n}.\end{aligned} \tag{A-10}$$

The TFP misallocation statistic $\mathcal{M}$ is then given by

$$\begin{aligned}\mathcal{M} =& \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int\left[\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}\right]dm}}{\Lambda} \\ =& \frac{e^{\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\left[S_{\varphi_k,\varphi_n}\left(\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\right)-\frac{1}{\theta}\right)\right]}}{\Lambda},\end{aligned}$$

where I have made use of

$$\int_1^{\infty}\ln\left(\frac{1}{m}\right)S_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm = \left[\frac{\theta\ln(m)+1}{\theta m^{\theta}}+C\right]_1^{\infty} = -\frac{1}{\theta}.$$

[26]For the derivation, I assume a continuous distribution of quality gaps.

Alternatively note that this expression is equal to $-S_{\varphi_k,\varphi_n}E[\ln(\lambda^{\Delta})|\varphi_k,\varphi_n]$. I have shown above that $\ln(\lambda^{\Delta})$ conditional on the productivity gap is exponentially distributed with parameter θ. From the characteristics of an exponential distribution, its expected value is $1/\theta$.

The aggregate markup is then given by

$$
\begin{aligned}
E[\mu] =& \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{\varphi_k}{\varphi_n}mS_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\
=& \frac{\theta}{\theta-1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{\varphi_k}{\varphi_n}S_{\varphi_k,\varphi_n}. \qquad \text{(A-11)}
\end{aligned}
$$

This concludes the derivations of the aggregate labor income share, TFP misallocation statistic and aggregate markup. In Peters (2020), the step size of innovation and the rate of creative destruction relative to internal R&D, captured by θ, fully characterize the aggregate labor income share, the misallocation measure, and the aggregate markup. In this model, these statistics further depend on the size and distribution of the productivity gaps. For example, a rise in the productivity gap or a reallocation of sales shares towards high-productivity firms lowers the aggregate labor income share in eq. (A-10) and raises the aggregate markup in eq. (A-11), ceteris paribus.

C.3 Deriving the steady-state growth rate of aggregate variables

The growth rate of Q_t determines the growth rate of aggregate variables

$$
g = \frac{\dot{Q}_t}{Q_t} = \frac{\partial\ln(Q_t)}{\partial t}.
$$

Quality of a product in a given product line increases through internal R&D, expansion R&D or firm entry. For the growth rate of Q_t over a discrete time interval Δ, we have

$$
\ln(Q_{t+\Delta}) = \int_0^1\Big[(\Delta I+\Delta Sx^h+\Delta(1-S)x^l+\Delta z)\ln(\lambda)+\ln(q_{t,i})\Big]di
$$

so that

$$
\frac{\ln(Q_{t+\Delta})-\ln(Q_t)}{\Delta} = \Big(I+Sx^h+(1-S)x^l+z\Big)\ln(\lambda).
$$

For $\Delta\to 0$, $g = \Big(I+Sx^h+(1-S)x^l+z\Big)\ln(\lambda)$ as stated in Proposition 1.

C.4 Solving for the steady state equilibrium

In the model, there are the seven unknown variables $x^h, x^l, I, z, \tau, \frac{Y_t}{w_t}, S$ and the markup distribution $\nu()$ in seven equations plus the system of differential equations characterizing $\nu()$.

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta - 1}{\psi_I} I^\zeta\right)\left(1 - \frac{1}{\lambda}\right)\frac{\psi_I}{\zeta(\rho + \tau)}\right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^h)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1}\frac{\varphi^l}{\varphi^h}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^l)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1}\frac{\varphi^h}{\varphi^l}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}$$

Free entry condition

$$p^h\Big(SV_t^h(1, \lambda) + (1 - S)V_t^h(1, \lambda \times \varphi^h/\varphi^l)\Big) + (1 - p^h)\Big(SV_t^l(1, \lambda \times \varphi^l/\varphi^h) + (1 - S)V_t^l(1, \lambda)\Big) = \frac{1}{\psi_z} w_t,$$

where

$$V_t^d(1, \mu) = \frac{1}{(\rho + \tau)}\frac{\zeta - 1}{\psi_x}(x^d)^\zeta w_t + \frac{Y_t\left(1 - \frac{1}{\mu}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta w_t \mu^{-1}}{\rho + \tau}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t}\sum_{\frac{\varphi_j}{\varphi_{j'}}}\sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}}\nu\left(i, \frac{\varphi_j}{\varphi_{j'}}\right) + \frac{1}{\psi_I} I^\zeta \sum_{\frac{\varphi_j}{\varphi_{j'}}}\sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}}\nu\left(i, \frac{\varphi_j}{\varphi_{j'}}\right) + S\frac{1}{\psi_x}(x^h)^\zeta + (1 - S)\frac{1}{\psi_x}(x^l)^\zeta + \frac{z}{\psi_z}$$

Creative destruction

$$\tau = z + Sx^h + (1 - S)x^l$$

Share of high productivity type

$$S = \sum_{i=1}^{\infty} \left[\nu\left(i, \frac{\varphi^h}{\varphi^h}\right) + \nu\left(i, \frac{\varphi^h}{\varphi^l}\right) \right],$$

where ν, the stationary distribution of quality and productivity gaps, is characterized by

$$0 = \dot{\nu}\left(\Delta, \frac{\varphi_j}{\varphi_{j'}}\right) = I\nu\left(\Delta - 1, \frac{\varphi_j}{\varphi_{j'}}\right) - \nu\left(\Delta, \frac{\varphi_j}{\varphi_{j'}}\right)(I + \tau) \quad \text{for} \quad \Delta \geq 2$$

and for the case of a unitary quality gap

$$\begin{aligned}
0 = \dot{\nu}\left(1, \frac{\varphi^l}{\varphi^h}\right) &= (1 - S)x^l S + z_t(1 - p^h)S - \nu\left(1, \frac{\varphi^l}{\varphi^h}\right)(I + \tau) \\
0 = \dot{\nu}\left(1, \frac{\varphi^l}{\varphi^l}\right) &= (1 - S)x^l(1 - S) + z_t(1 - p^h)(1 - S) - \nu\left(1, \frac{\varphi^l}{\varphi^l}\right)(I + \tau) \\
0 = \dot{\nu}\left(1, \frac{\varphi^h}{\varphi^h}\right) &= Sx^h S + z_t p^h S - \nu\left(1, \frac{\varphi^h}{\varphi^h}\right)(I + \tau) \\
0 = \dot{\nu}\left(1, \frac{\varphi^h}{\varphi^l}\right) &= Sx^h(1 - S) + z_t p^h(1 - S) - \nu\left(1, \frac{\varphi^h}{\varphi^l}\right)(I + \tau).
\end{aligned}$$

To simplify the system of equations, first rewrite the rate of creative destruction

$$z = (\tau - Sx^h - (1 - S)x^l)$$

such that z can be substituted out from the remaining equations. Second, based on Proposition 1, we know

$$S = \frac{Sx^h + zp^h}{\tau}.$$

Third, the optimality conditions for expansion rates (multiplied by p^h and $(1 - p^h)$) and the free entry condition together imply

$$\frac{1}{\psi_x} p^h (x^h)^{\zeta - 1} + \frac{1}{\psi_x}(1 - p^h)(x^l)^{\zeta - 1} = \frac{1}{\psi_z \zeta}.$$

The system of equilibrium conditions can hence be reduced to:

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t}\psi_I - (\zeta - 1) I^\zeta \right) \frac{\left(1 - \frac{1}{\lambda}\right)}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^h)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{1}{\psi_I}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta\lambda^{-1}\frac{\varphi^l}{\psi_I\varphi^h}\right) = (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^l)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + (\zeta - 1)I^\zeta \lambda^{-1}\frac{\varphi^h}{\psi_I\varphi^l}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^\zeta\lambda^{-1}\frac{1}{\psi_I}\right) = (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}$$

Free entry

$$p^h\frac{(x^h)^{\zeta - 1}}{\psi_x} + (1 - p^h)\frac{(x^l)^{\zeta - 1}}{\psi_x} = \frac{1}{\psi_z\zeta}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t}\Lambda + \Lambda_I + S\frac{1}{\psi_x}(x^h)^\zeta + (1 - S)\frac{1}{\psi_x}(x^l)^\zeta + \frac{\tau - Sx^h - (1 - S)x^l}{\psi_z},$$

where

$$\begin{aligned}
\Lambda =& \frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n} \\
\Lambda_I =& \frac{1}{\psi_I}I^\zeta\frac{\theta}{\theta + 1}\sum_{k \in \{h,l\}}\sum_{n \in \{h,l\}}\frac{1}{\varphi_k/\varphi_n}S_{\varphi_k,\varphi_n} \\
\theta =& \frac{\ln(I + \tau) - \ln(I)}{\ln(\lambda)}
\end{aligned}$$

Sales share of high productivity type

$$S = \frac{Sx^h + (\tau - Sx^h - (1 - S)x^l)p^h}{\tau}$$

The expressions related to the labor market clearing condition are derived in Section C.2. This constitutes a system of seven equations in seven unknowns $(x^h, x^l, I, \tau, \frac{Y_t}{w_t}, S)$, which I solve using a root finder.

C.5 Firm markup growth

I show in the Appendix, Section C.6 that the expected log markup conditional on firm age a_f for a high-productivity type firm is

$$E\left[\ln \mu_f | a_f, \varphi^h\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^h | a_f]\right)}_{\text{Quality improvements}} + \underbrace{(1 - S) \times \ln\left(\frac{\varphi^h}{\varphi^l}\right)}_{\text{Productivity advantage}}, \tag{A-12}$$

where $E[a_P^h|a_f]$ is the average product age of a high-type firm conditional on firm age.

The expected firm markup conditional on age consists of two terms. The first term in eq. (A-12) is akin to Peters (2020) and reflects that internal R&D at the product level translates into markup growth at the firm level as it ages. In Peters (2020), this term holds for all firms, whereas in this model, this term is specific to the productivity type of the firm, as the average product age varies by firm type. The second term in eq. (A-12) captures a new level effect that heterogeneity in productivity introduces. The intuition is that if a high-type incumbent faces a low-type second-best firm, it can charge a φ^h/φ^l higher markup, which occurs in expectation in $1 - S$ of the incumbent's product lines. Expected markup growth conditional on survival then equals $E\left[\ln \mu_f | a_f, \varphi^h\right] - E\left[\ln \mu_f | 0, \varphi^h\right]$.

The expected markup conditional on firm age for a low-productivity type firm follows

$$E\left[\ln \mu_f | a_f, \varphi^l\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^l | a_f]\right)}_{\text{Quality improvements}} + \underbrace{S \times \ln\left(\frac{\varphi^l}{\varphi^h}\right)}_{\text{Productivity disadvantage}}. \tag{A-13}$$

The first term captures quality improvements through internal R&D. The second term in eq. (A-13) differs from eq. (A-12). Low-productivity incumbents face a high-productivity second-best firm in a share S of their product lines. Since $\varphi^l < \varphi^h$, this term is negative.

C.6 Firm markups

Firm markups are defined by $\mu_f = \frac{p y_f}{w l_f} = \left(\frac{1}{n}\sum_{k=1}^{n} \mu_{kf}^{-1}\right)^{-1}$. Therefore

$$\ln \mu_f = -\ln\left(\frac{1}{n}\sum_{k=1}^{n} \mu_k^{-1}\right).$$

Rewrite the term in brackets (for a high-productivity firm) as

$$\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1}=\frac{1}{n}\sum_{k=1}^{n}e^{-\ln\mu_k}=\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda}+\sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right), \tag{A-14}$$

where i indexes the product lines where the high productivity firm faces a low productivity second best producer, j the lines where it faces a high productivity second best producer and $n_i+n_j=n$. A two-dimensional linear Taylor expansion around $\ln\lambda=0$ and $\ln\frac{\varphi^h}{\varphi^l}=0$ gives

$$\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda}+\sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right)\approx 1-\left(\frac{1}{n}\sum_{k=1}^{n}\Delta_k\right)\ln\lambda-\frac{n_i}{n}\ln\left(\frac{\varphi^h}{\varphi^l}\right)$$

such that

$$E\left[\ln\mu_f|\text{firm age}=a_f,\varphi^h\right]\approx E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age}=a_f,\varphi^h\right]\ln\lambda+(1-S)\ln\left(\frac{\varphi^h}{\varphi^l}\right),$$

where I have used the fact that (in expectation) the share of the firm's product lines with a low productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the low productivity type. From Peters (2020), we know that

$$E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age}=a_f,\varphi^h\right]\ln\lambda=\left(1+I\times E[a_P^h|a_f]\right)\ln\lambda,$$

where $E[a_P^h|a_f]$ denotes the average product age of a high-productivity type firm conditional on firm age a_f and

$$\begin{aligned}
E[a_P^h|a_f]&=\frac{1}{x^h}\left(\frac{\frac{1}{\tau}\left(1-e^{-\tau a_f}\right)}{\frac{1}{x^h+\tau}\left(1-e^{-(x^h+\tau)a_f}\right)}-1\right)\left(1-\phi^h(a_f)\right)+a_f\phi^h(a_f)\\
\phi^h(a)&=e^{-x^h a}\frac{1}{\gamma^h(a)}\ln\left(\frac{1}{1-\gamma^h(a)}\right)\\
\gamma^h(a)&=\frac{x^h\left(1-e^{-(\tau-x^h)a}\right)}{\tau-x^h e^{-(\tau-x^h)a}}.
\end{aligned}$$

For a firm of the low-productivity type, the last term in eq. (A-14) reads

$$\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\Delta_i\ln\lambda}+\sum_{j=1}^{n_j}e^{-\ln\frac{\varphi^l}{\varphi^h}-\ln\Delta_j\ln\lambda}\right),$$

where i indexes the product lines where the low-productivity producer faces a low-productivity second best producer, j the lines where it faces a high-productivity second best producer and $n_i + n_j = n$. Following the same steps as for a high-productivity firm, this time linearizing around $\ln \frac{\varphi^l}{\varphi^h} = 0$ (and $\ln \lambda = 0$) gives

$$E\left[\ln \mu_f | \text{firm age} = a_f, \varphi^l\right] \approx \left(1 + I \times E[a_P^l | a_f]\right) \ln \lambda + S \ln \left(\frac{\varphi^l}{\varphi^h}\right),$$

where again I have made use of the fact that (in expectation) the share of the firm's product lines with a high-productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the high-productivity type. $E[a_P^l | a_f]$ is exactly defined as $E[a_P^h | a_f]$ with x^h replaced by x^l in the above expressions.

C.7 Firm size distribution

The mass of high- and low-productivity type firms with $n \geq 2$ products follows the differential equations

$$\begin{aligned} \dot{M}_t^h(n) &= (n-1)x_t^h M_t^h(n-1) + (n+1)\tau_t M_t^h(n+1) - n(x_t^h + \tau_t)M_t^h(n) \\ \dot{M}_t^l(n) &= (n-1)x_t^l M_t^l(n-1) + (n+1)\tau_t M_t^l(n+1) - n(x_t^l + \tau_t)M_t^l(n), \end{aligned} \tag{A-15}$$

whereas the mass of firms with one product evolves according to

$$\begin{aligned} \dot{M}_t^h(1) &= z_t p^h + 2\tau_t M_t^h(2) - (x_t^h + \tau_t)M_t^h(1) \\ \dot{M}_t^l(1) &= z_t(1 - p^h) + 2\tau_t M_t^l(2) - (x_t^l + \tau_t)M_t^l(1). \end{aligned} \tag{A-16}$$

The mass of firms with n products increases through firms with $n-1$ products expanding to size n at rate x_t^h or x_t^l per product or through firms with $n+1$ products losing a product at the rate of aggregate creative destruction τ_t. The mass of firms with n products decreases through firms with n products either gaining or losing a product through expansion or creative destruction. The mass of firms with one product additionally increases through firm entry.

Proposition A-1. *The stationary firm size distribution along the balanced growth path is characterized as follows.*

1. *The mass of high and low productivity firms with n products is*

$$M^h(n) = \frac{(x^h)^{n-1} z p^h}{n\tau^n} = \frac{zp^h}{x^h}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n$$
$$M^l(n) = \frac{(x^l)^{n-1} z(1-p^h)}{n\tau^n} = \frac{z(1-p^h)}{x^l}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n.$$

2. *The total mass of firms with n products is*

$$M(n) = M^h(n) + M^l(n) = \frac{(x^h)^{n-1} z p^h + (x^l)^{n-1} z(1-p^h)}{n\tau^n}.$$

3. *The mass of firms of each productivity type is*

$$M^h = \sum_{n=1}^{\infty} M^h(n) = \frac{zp^h}{x^h}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n = \frac{zp^h}{x^h}\ln\left(\frac{\tau}{\tau - x^h}\right)$$
$$M^l = \sum_{n=1}^{\infty} M^l(n) = \frac{z(1-p^h)}{x^l}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n = \frac{z(1-p^h)}{x^l}\ln\left(\frac{\tau}{\tau - x^l}\right)$$

4. *The total mass of firms is*

$$M = M^h + M^l.$$

Proof. These results follow from setting the time derivatives in equations (A-15) and (A-16) equal to zero and solving the system of equations. □

For each firm type, the share of firms with n products, $M^h(n)/M^h$ and $M^l(n)/M^l$, follows the PDF of a logarithmic distribution with parameter x^h/τ and x^l/τ as in Lentz and Mortensen (2008). The firm size distribution is highly skewed to the right.

Since there is a continuum of mass one of products and each product is mapped to one firm $\sum_{i=1}^{\infty} M(n) \times n = 1$. Further, the mass of high-productivity type firms producing n products is related to the share of product lines operated by high-type firms, S, as follows

$$S = \sum_{n=1}^{\infty} M^h(n) \times n = \frac{zp^h}{\tau - x^h}.$$

D Computation of transition dynamics

In this section, I lay out the numerical procedure to solve for the transition path. Since time is continuous, I solve a discretized version of the model where the solution converges to the one in continuous time for small enough time intervals. As shown in Appendix C, value functions are additive across product lines. Therefore, I solve the problem of two representative one-product firms: one of the high productivity type and one of the low productivity type.

I normalize the value function by the wage w_t to obtain a stationary problem. The value function for the high-type firm (in discrete time) reads

$$
\begin{aligned}
\frac{V_t^h(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^h}\left\{x_t^h \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^h(1,\lambda,S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^h(1,\lambda\frac{\varphi^h}{\varphi^l},S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^h)^\zeta dt\right\} \\
&+ \max_{I_t^h}\left\{I_t^h \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^h(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-17}
$$

The value function for the low-type firm reads

$$
\begin{aligned}
\frac{V_t^l(1,\mu_i,S_t)}{w_t} &= \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^l}\left\{x_t^l \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^l(1,\lambda\frac{\varphi^l}{\varphi^h},S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^l(1,\lambda,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^l)^\zeta dt\right\} \\
&+ \max_{I_t^l}\left\{I_t^l \exp(-r_{t+dt}dt)\left(\frac{V_{t+dt}^l(1,\mu_i\lambda,S_{t+dt})}{w_{t+dt}} - \frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_I}\mu_i^{-1}(I_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^l(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned} \tag{A-18}
$$

From this, one obtains the first order conditions for the policy functions. For the optimal expansion R&D rate of the high type firm x_t^h (again suppressing the dependence of the value function on S_t):

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^h_{t+dt}(1,\lambda)}{w_{t+dt}}+(1-S_{t+dt})\frac{V^h_{t+dt}(1,\lambda\frac{\varphi^h}{\varphi^l})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^h_t)^{\zeta-1} \tag{A-19}$$

and for the low type firm x^l_t:

$$\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V^l_{t+dt}(1,\lambda\frac{\varphi^l}{\varphi^h})}{w_{t+dt}}+(1-S_{t+dt})\frac{V^l_{t+dt}(1,\lambda)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_x}(x^l_t)^{\zeta-1}. \tag{A-20}$$

Both are independent of the markup μ_i. For the optimal internal R&D rates of the high type, I^h_t, one obtains

$$\exp(-r_{t+dt}dt)\left(\frac{V^h_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^h_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^h_t)^{\zeta-1} \tag{A-21}$$

and similarly for I^l_t

$$\exp(-r_{t+dt}dt)\left(\frac{V^l_{t+dt}(1,\mu_i\lambda)}{w_{t+dt}}-\frac{V^l_{t+dt}(1,\mu_i)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}=\frac{\zeta}{\psi_I}\mu_i^{-1}(I^l_t)^{\zeta-1}. \tag{A-22}$$

Equations (A-17) to (A-22) characterize the firm problem in discrete time. These equations are supplemented by the law of motion for the two dimensional distribution of quality and productivity gaps

$$\nu_{t+dt}\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)=dt\left[I_{\mu_i,t}\nu_t\left(\Delta-1,\frac{\varphi_j}{\varphi_{j'}}\right)-\nu_t\left(\Delta,\frac{\varphi_j}{\varphi_{j'}}\right)(I_{\mu_i,t}+\tau_t)\right]\quad\text{for}\quad\Delta\geq 2 \tag{A-23}$$

and for product lines with a unitary quality gap, $\Delta=1$,

$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)=dt\left[(1-S_t)x^l_tS_t+z_t(1-p^h)S_t-\nu_t\left(1,\frac{\varphi^l}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^l}{\varphi^l}\right)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)=dt\left[(1-S_t)x^l_t(1-S_t)+z_t(1-p^h)(1-S_t)-\nu_t\left(1,\frac{\varphi^l}{\varphi^l}\right)(I_{\mu_i,t}+\tau_t)\right]$$
$$\nu_{t+dt}\left(1,\frac{\varphi^h}{\varphi^h}\right)-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)=dt\left[S_tx^h_tS_t+z_tp^hS_t-\nu_t\left(1,\frac{\varphi^h}{\varphi^h}\right)(I_{\mu_i,t}+\tau_t)\right]$$

$$\nu_{t+dt}\left(1,\frac{\varphi^h}{\varphi^l}\right)-\nu_t\left(1,\frac{\varphi^h}{\varphi^l}\right)=dt\left[S_t x_t^h(1-S_t)+z_t p^h(1-S_t)-\nu_t\left(1,\frac{\varphi^h}{\varphi^l}\right)(I_{\mu_i,t}+\tau_t)\right] \tag{A-24}$$

and a standard Euler equation

$$\frac{C_{t+dt}}{C_t}=\exp(-\rho dt)(1+r_{t+dt}dt). \tag{A-25}$$

Further, the (static) free entry and labor market clearing conditions remain unchanged and are characterized in the main text by equations (10) and (12).

The algorithm to compute the transition path assumes that an initial and ending balanced growth path have been solved for including the (stationary) two-dimensional distribution of quality and productivity gaps. I choose $dt = 0.02$ and set the transition period to 300 years (T), after which I assume the economy has reached its new balanced growth path. I further truncate the two dimensional distribution of quality and productivity gaps along the quality dimension at $\Delta = 30$, implying a maximum quality gap of λ^{30}. No mass reaches this state during the transition such that this assumption is satisfied. I then compute the transition path as follows:

1. Guess a path of wage growth $\frac{w_{d+dt}}{w_t}$ and income to wage ratios Y_t/w_t over the transition (equal to their values in the final balanced growth path)

   (a) Guess a path for S_t over the transition (equal to its value in the final balanced growth path).

      i. Starting backwards in period T, solve for optimal policy functions in $T - dt$ using equations (A-19)-(A-22).[27]

      ii. Solve for τ_{T-dt} that ensures that the free entry condition (10) holds.

      iii. Compute the value function in $T - dt$ using equations (A-17) and (A-18).

      iv. Iterate backwards until the first time period.

[27] I solve for the optimal I_i (at each point in time) over a two-dimensional grid of quality and productivity gaps.

v. Starting from the initial balanced growth path, simulate S_t forward using[28]

$$S_{t+dt} = S_t + dt\left[S_t x_t^h(1-S_t) - (1-S_t)x_t^l S_t + z_t(p^h(1-S_t) - (1-p^h)S_t)\right],$$

where z_t can be substituted out by equation (9).

(b) Update the guess for S_t from step v using bisection and go back to step i. Iterate until the guessed path for S_t converges to the implied one.

2. Starting from the initial balanced growth path, simulate the two dimensional distribution of quality and productivity gaps forward using equations A-23 and A-24.
3. Solve for the implied sequence of $\frac{Y_t}{w_t}$ from the labor market clearing condition.
4. Compute the sequence of quality growth using

$$\frac{Q_{t+dt}}{Q_t} = \exp\left(\left[\int_0^1 I_{\mu_i,t+dt}di + S_{t+dt}x_{t+dt}^h + (1-S_{t+dt})x_{t+dt}^l + z_{t+dt}\right] dt\ln(\lambda)\right).$$

5. Compute the sequence of aggregate productivity growth using

$$\frac{\Phi_{t+dt}}{\Phi_t} = \left(\frac{\varphi^h}{\varphi^l}\right)^{S_{t+dt}-S_t}.$$

6. Using the two dimensional distribution of quality and productivity gaps, compute the sequence of $\mathcal{M}_t$ defined in equation (8).
7. Compute the sequence of production labor L_{Pt} using equation (6).
8. Compute the sequence of aggregate output growth $\frac{Y_{t+dt}}{Y_t}$ using equation (8).
9. With the paths of aggregate output growth and $\frac{Y_t}{w_t}$, obtain the implied path of wage growth $\frac{w_{d+dt}}{w_t}$.
10. Update the guesses for $\frac{w_{d+dt}}{w_t}$ and $\frac{Y_t}{w_t}$ using bisection and go back to step (a). Iterate until the guessed and implied paths converge.

I declare convergence when the difference between the three guessed and implied sequences is less than $\epsilon = 10^{-5}$ in absolute terms. I annualize wage growth before applying the convergence threshold.

[28]One could already simulate the entire two-dimensional distribution of quality and productivity gaps forward here. However, for the inner loop, it is sufficient to iterate on S_t.

E Alternative decomposition

I decompose the contribution of each parameter change to the change in the targeted moments across the initial and new balanced growth path as follows

$$\frac{M_j\left(\boldsymbol{\theta}_{-i}^{new}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{new}, \theta_i^{initial}\right)}{M_j\left(\boldsymbol{\theta}_{-i}^{new}, \theta_i^{new}\right) - M_j\left(\boldsymbol{\theta}_{-i}^{initial}, \theta_i^{initial}\right)} \times 100, \tag{A-26}$$

where *initial* and *new* index the initial and new balanced growth path.[29] The difference to the decomposition in the main text is that eq. (A-26) quantifies the contribution of each parameter change starting from the new balanced growth path. Table A-3 shows the contribution of each parameter change (rows) to the five moments of the estimation (columns). The notation follows Table 3.

Table A-3: Alternative decomposition

	VA	wL	σ_{VA}	$\frac{z}{M}$	g
Parameter Δ					
Expansion R&D efficiency $\psi_x \uparrow$	240.8	228.2	133.7	219.3	-5.8
Internal R&D efficiency $\psi_I \uparrow$	-20.3	-20.2	-100.0	-123.1	-23.2
Productivity gap $\varphi^h/\varphi^\ell \uparrow$	29.4	26.8	61.5	94.4	8.4
Entry R&D efficiency $\psi_z \uparrow$	-627.3	-580.0	-2276.4	-2711.9	-170.3
Step size of quality improvements $\lambda \downarrow$	71.1	68.5	116.9	422.3	256.5

Notes: the table shows the contribution of each parameter change (rows) to the changes in the five moments (columns) across the intial and new balanced growth path (BGP) in percent. The contribution of each parameter change is computed using eq. (A-26). Moments and parameter values of the initial and new BGP are shown in Table 4. VA denotes the difference in average log value added between firms aged eight and entrants, wL the difference in the average log wage bill between firms aged eight and entrants, σ_{VA} the standard deviation of log value added across firms, $\frac{z}{M}$ the entry rate and g the aggregate growth rate.

Table A-3 confirms the results of the decomposition in the main text. The increase in firm size relative to entry and the rise in firm-size dispersion are mainly driven by the increase in the expansion R&D efficiency, whereas the fall in the step size of quality improvements accounts for the fall in productivity growth. The decomposition in Table A-3 attributes a somewhat larger contribution to the fall in the step size of quality improvements than the one in the main text, however, the role of each parameter change is the same as in the previous decomposition. The model becomes highly-nonlinear for low levels of firm entry, which explains the increased contribution of ψ_z in Table A-3.

[29] Setting ψ_I or ψ_z to their values in the initial balanced growth path with all other parameters set at their new balanced growth path values leads to zero firm entry. Without entry, the firm-size distribution is non-stationary. Hence, for ψ_I or ψ_z, I compute the contribution by multiplying the contribution of a 0.1% change of the total change in the respective parameter by a factor of 1000.